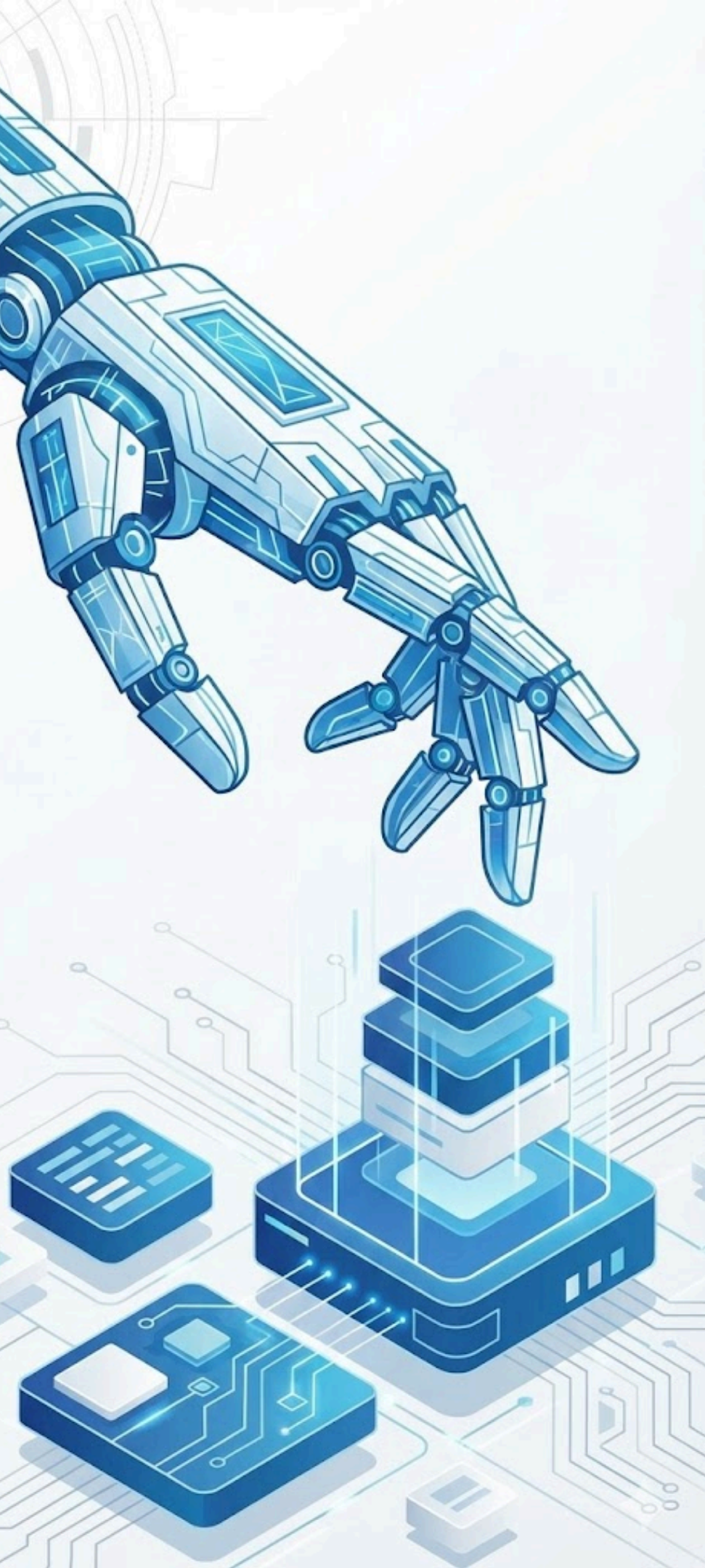


Ros2 Cpp 환경에서 PointCloud 센서 데이터 찍먹해보기

바질컴퍼니 AI Robotics Researcher 김병찬



목차

1. 로컬 환경 구축

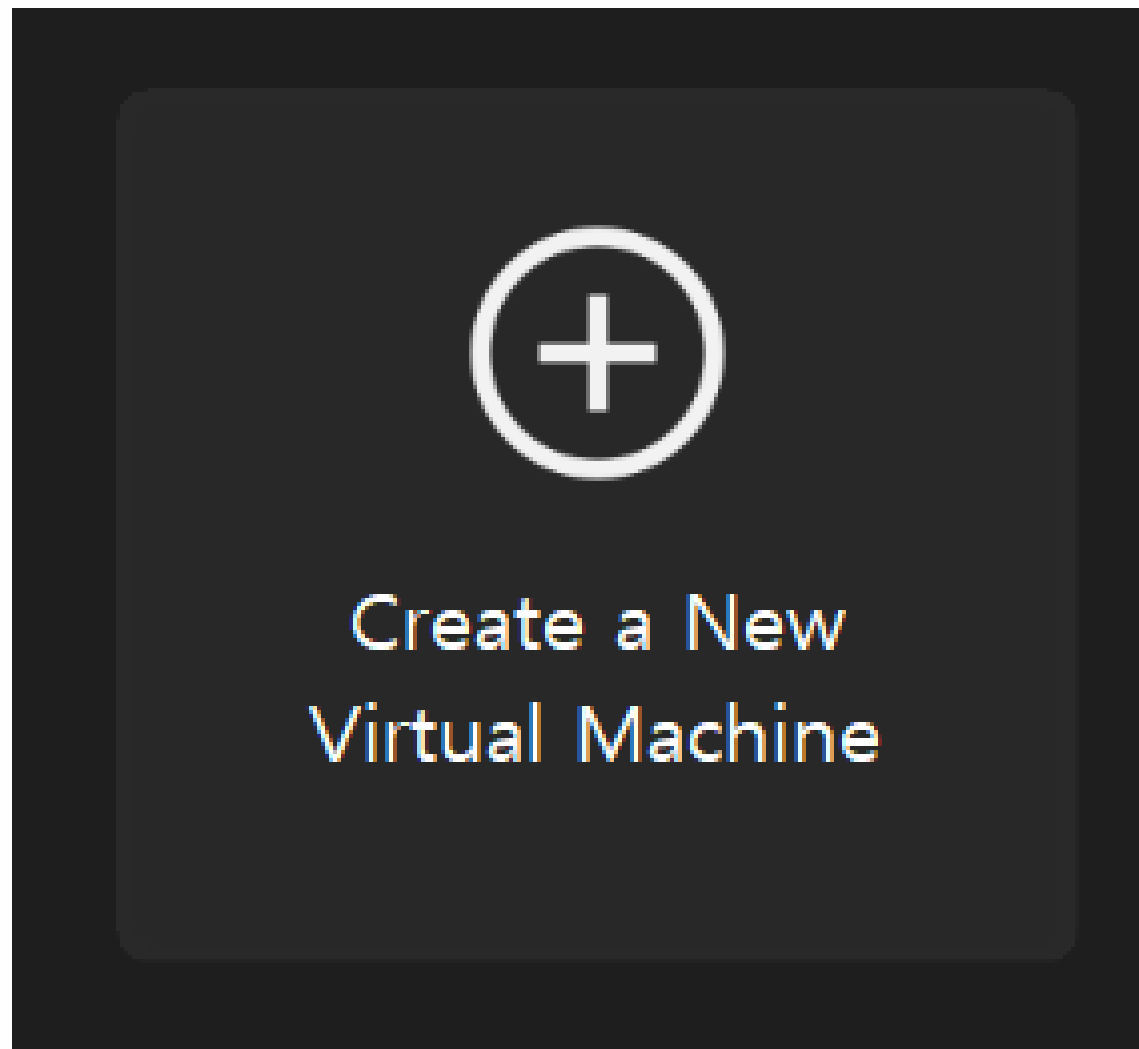
2. Cpp

3. Ros2 cpp

4. Pcl 데이터 찍먹

1. 로컬 환경 구축

먼저 VMWare Pro를 켜주세요!



Chrome에 **Ubuntu 22.04 iso**를 검색해서 설치해주세요

Select an image

Ubuntu is distributed on three types of images described below.

Desktop image

The desktop image allows you to try Ubuntu without changing your computer at all, and at your option to install it permanently later. This type of image is what most people will want to use. You will need at least 1024MiB of RAM to install from this image.

64-bit PC (AMD64) desktop image

Choose this if you have a computer based on the AMD64 or EM64T architecture (e.g., Athlon64, Opteron, EM64T Xeon, Core 2). Choose this if you are at all unsure.

☒ **Installer disc image file (iso)**

C:\Users\LEGION\Desktop\Iso images\ubuntu-22. ✓

Browse...

New Virtual Machine Wizard



Easy Install Information

This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

User name:

Password:

Confirm:

Help

< Back

Next >

Cancel

New Virtual Machine Wizard



Ready to Create Virtual Machine

Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.

The virtual machine will be created with the following settings:

Name:	Ubuntu 64-bit
Location:	C:\Users\LEGION\Documents\Virtual Machines\Ubun...
Version:	Workstation 25H2
Operating System:	Ubuntu 64-bit
Hard Disk:	20 GB, Split
Memory:	4096 MB
Network Adapter:	NAT
Other Devices:	2 CPU cores, CD/DVD, USB Controller, Sound Card

[Customize Hardware...](#)

☒ [Power on this virtual machine after creation](#)

< [Back](#)

[Finish](#)

[Cancel](#)

Install

Keyboard layout

Choose your keyboard layout:

English (Australian)
English (Cameroon)
English (Ghana)
English (Nigeria)
English (South Africa)
English (UK)
English (US)
Esperanto
Estonian
Faroese
Filipino
Finnish
French
French (Canada)

English (US)
English (US) - Cherokee
English (US) - English (Colemak)
English (US) - English (Colemak-DH ISO)
English (US) - English (Colemak-DH)
English (US) - English (Dvorak)
English (US) - English (Dvorak, alt. intl.)
English (US) - English (Dvorak, intl., with dead keys)
English (US) - English (Dvorak, left-handed)
English (US) - English (Dvorak, right-handed)
English (US) - English (Macintosh)
English (US) - English (Norman)
English (US) - English (US, Symbolic)
English (US) - English (US, intl.)

Type here to test your keyboard

Detect Keyboard Layout

Quit

Back

Continue

> Verifying the installation configuration...

Updates and other software

What apps would you like to install to start with?

☐ Normal installation

Web browser, utilities, office software, games, and media players.

☒ Minimal installation

Web browser and basic utilities.

Other options

☒ Download updates while installing Ubuntu

This saves time after installation.

☐ Install third-party software for graphics and Wi-Fi hardware and additional media formats

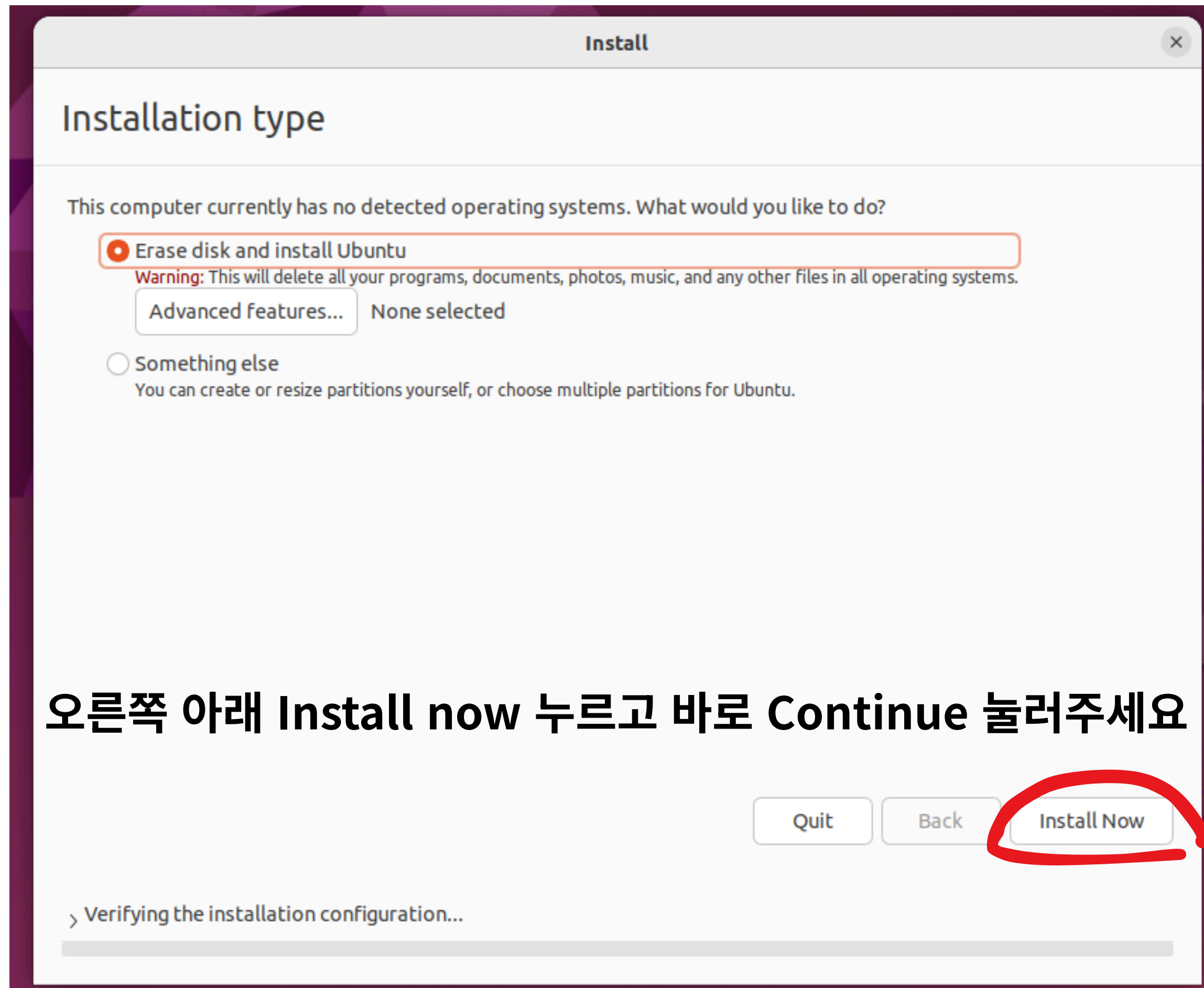
This software is subject to license terms included with its documentation. Some is proprietary.

Quit

Back

Continue

> Verifying the installation configuration...



오른쪽 아래 Install now 누르고 바로 Continue 눌러주세요

Install

Where are you?



Seoul

Back

Continue

› Creating ext4 file system for / in partition #3 of SCSI33 (0,0,0) (sda)...



Install

Who are you?

Your name:

bssm



Your computer's name:

bssm-virtual-machine



The name it uses when it talks to other computers.

Pick a username:

bssm



Choose a password:

1234



Short password

Confirm your password:

1234



☐ Log in automatically

☒ Require my password to log in

☐ Use Active Directory

You'll enter domain and other details in the next step.

Back

Continue

> Almost finished copying files...



이제 Ubuntu 설치가 시작되는데 좀 걸려요

다음 링크를 Chrome 에 검색해서 들어와주세요!

<https://github.com/Bchan725>

BSSM-winter-camp-PCL

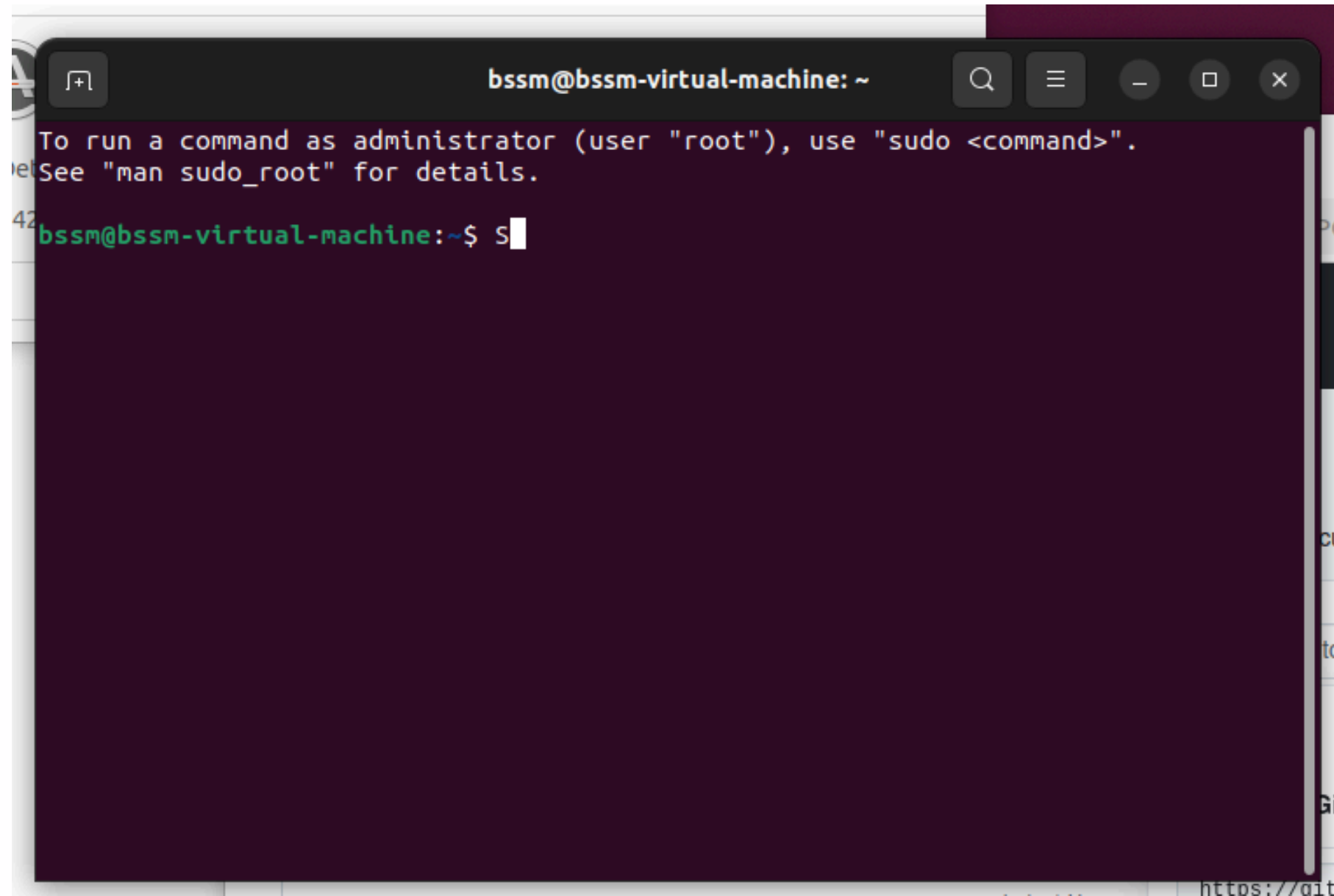
Public



C++

Updated 5 hours ago

터미널 실행 : Control + Alt + T



A screenshot of a terminal window titled "bssm@bssm-virtual-machine: ~". The window has a dark background and a light-colored title bar. The terminal content shows a message: "To run a command as administrator (user \"root\"), use \"sudo <command>\". See \"man sudo_root\" for details." followed by a prompt "bssm@bssm-virtual-machine:~\$". The user has entered the letter "S" and the cursor is at the end of the command line. The terminal window is overlaid on a desktop background that is partially visible on the right side.

```
bssm@bssm-virtual-machine: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
bssm@bssm-virtual-machine:~$ S
```

Repo Clone

Copy : Control + Shift + C

Paste : Control + Shift + v

```
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$ code .
Command 'code' not found, but can be installed with:
sudo snap install code
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$
```

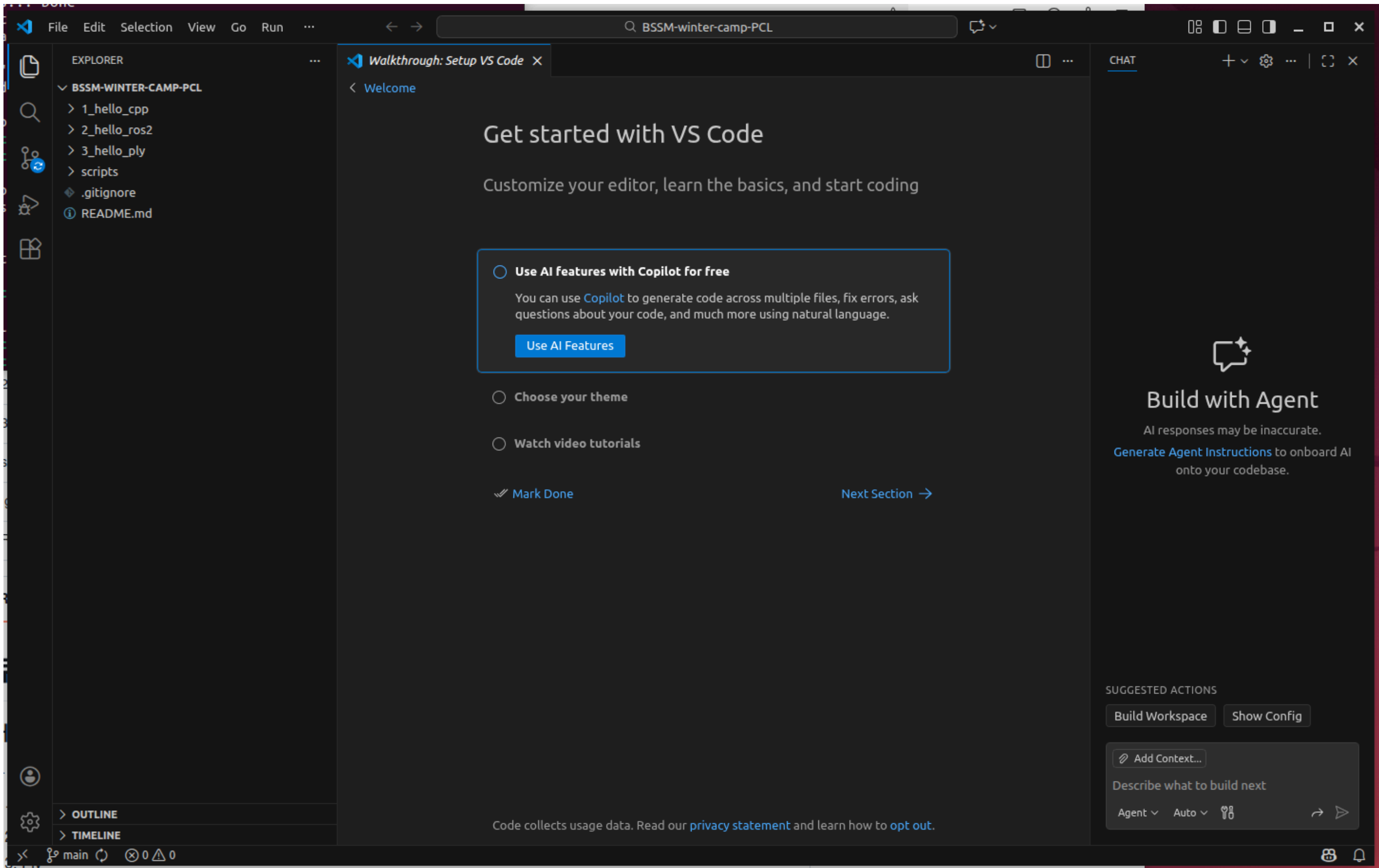
sudo apt update
sudo snap install code --classic

```
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL
Building dependency tree... Done
Reading state information... Done
276 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

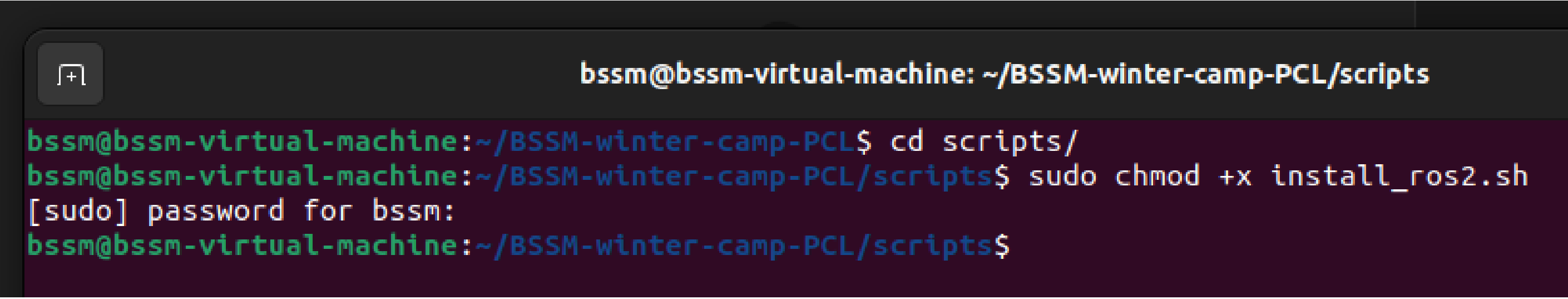
No apt package "code", but there is a snap with that name.
Try "snap install code"

E: Unable to locate package code
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$ sudo apt ^C
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$ sudo snap install code
error: This revision of snap "code" was published using classic confinement and
thus may perform arbitrary system changes outside of the security
sandbox that snaps are usually confined to, which may put your system at
risk.

If you understand and want to proceed repeat the command including
--classic.
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$ sudo snap install code --class
ic
code 994fd12f from Visual Studio Code (vscode✓) installed
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$
```



sudo chmod +x install_ros2.sh



```
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/scripts  
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$ cd scripts/  
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/scripts$ sudo chmod +x install_ros2.sh  
[sudo] password for bssm:  
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/scripts$
```

./install_ros2.sh

```
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/scripts
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/scripts$ cd scripts/
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/scripts$ sudo chmod +x install_ros2.sh
[sudo] password for bssm:
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/scripts$ ./install_ros2.sh
Get:1 http://kr.archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:2 http://kr.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://kr.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://kr.archive.ubuntu.com/ubuntu jammy/main i386 Packages [1,040 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:6 http://kr.archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1,395 kB]
Get:7 http://kr.archive.ubuntu.com/ubuntu jammy/main Translation-en [510 kB]
Get:8 http://kr.archive.ubuntu.com/ubuntu jammy/main amd64 DEP-11 Metadata [423 kB]
Get:9 http://kr.archive.ubuntu.com/ubuntu jammy/main DEP-11 48x48 Icons [100.0 kB]
Get:10 http://kr.archive.ubuntu.com/ubuntu jammy/main DEP-11 64x64 Icons [148 kB]
Get:11 http://kr.archive.ubuntu.com/ubuntu jammy/main DEP-11 64x64@2 Icons [15.8 kB]
Get:12 http://kr.archive.ubuntu.com/ubuntu jammy/main amd64 c-n-f Metadata [30.3 kB]
Get:13 http://kr.archive.ubuntu.com/ubuntu jammy/restricted i386 Packages [30.4 kB]
Get:14 http://kr.archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [129 kB]
Get:15 http://kr.archive.ubuntu.com/ubuntu jammy/restricted Translation-en [18.6 kB]
Get:16 http://kr.archive.ubuntu.com/ubuntu jammy/restricted amd64 c-n-f Metadata [488 B]
Get:17 http://kr.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
15% [17 Packages 1,014 kB/14.1 MB 7%]
```

```
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$ ros2
usage: ros2 [-h] [--use-python-default-buffering] Call 'ros2 <command> -h' for more detailed usage. ...

ros2 is an extensible command-line tool for ROS 2.

options:
  -h, --help                show this help message and exit
  --use-python-default-buffering
                           Do not force line buffering in stdout and instead use the python default buffering,
                           which might be affected by PYTHONUNBUFFERED/-u and depends on whatever stdout is
                           interactive or not

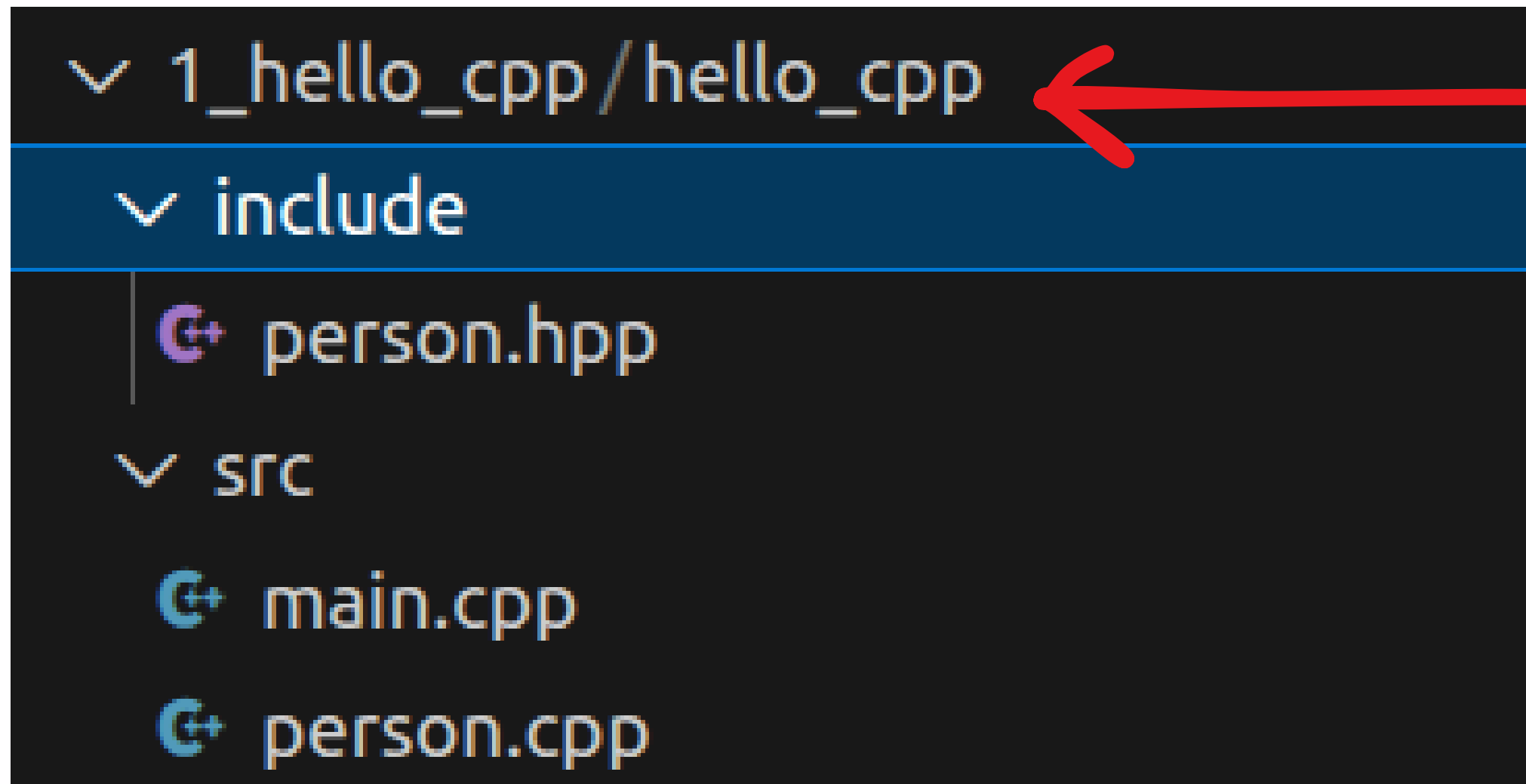
Commands:
  action                    Various action related sub-commands
  bag                      Various rosbag related sub-commands
  component                Various component related sub-commands
  daemon                   Various daemon related sub-commands
  doctor                   Check ROS setup and other potential issues
  interface                Show information about ROS interfaces
  launch                   Run a launch file
  lifecycle                Various lifecycle related sub-commands
  multicast                Various multicast related sub-commands
  node                     Various node related sub-commands
  param                   Various param related sub-commands
  pkg                      Various package related sub-commands
  run                      Run a package specific executable
  security                 Various security related sub-commands
  service                  Various service related sub-commands
  topic                   Various topic related sub-commands
  wtf                      Use 'wtf' as alias to 'doctor'

  Call 'ros2 <command> -h' for more detailed usage.
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$
```

2. Cpp

2. Cpp

hello cpp



Cpp 실습 폴더 위치

2. Cpp

코드 구조 설명 - main.cpp

#include 는 다들 써봐서 아실거라고 생각합니다??

Person 객체는 직접 생성되었음.

include/ : 안에 구조가 정의 되어있음

→ **person.hpp**

src/ : 안에 코드가 정의 되어있음.

→ **person.cpp**

```
#include "person.hpp"

#include <iostream>
#include <vector>

void space() {
    std::cout << std::endl;
}

int main()
{
    // 1. 객체 생성
    Person student1("홍길동", 20);

    // 2. 정보 출력
    student1.print_info();

    // 3. 취미 추가
    student1.add_hobby("축구");
    student1.add_hobby("독서");

    // 4. 취미 출력
    student1.print_hobbies();
}
```

person.hpp

1_hello_cpp > hello_cpp > include > person.hpp

```
1  #pragma once
2
3  #include <string>
4  #include <vector>
5
6  class Person
7  {
8  public:
9      Person(const std::string& name, int age);
10     ~Person();
11
12 public:
13     void print_info();
14     void print_hobbies();
15
16     void add_hobby(const std::string &hobby);
17
18 private:
19     std::string name_;
20     int age_;
21     std::vector<std::string> hobbies_;
22 };
23
```

멤버 변수

person.hpp

person.cpp

main.cpp

1_hello_cpp > hello_cpp > src > ~/BSSM-winter-camp-PCL/1_hello_cpp/hello_cpp/src/person.cpp

```
1  #include "person.hpp"
2
3  #include <string>
4  #include <vector>
5  #include <iostream>
6
7  Person::Person(const std::string& name, int age)
8      : name_(name)
9      , age_(age)
10 {}
11
12 Person::~Person() {}
13
14 void Person::print_info() {
15     std::cout << "이름: " << name_ << ", 나이: " << age_ << "세" <<
16 }
17
18 void Person::print_hobbies() {
19     std::cout << name_ << "의 취미: ";
20     for (size_t i = 0; i < hobbies_.size(); ++i) {
21         std::cout << hobbies_[i];
22         if (i < hobbies_.size() - 1) std::cout << ", ";
23     }
24     std::cout << std::endl;
25 }
26
27 void Person::add_hobby(const std::string& hobby) {
28     hobbies_.push_back(hobby);
29 }
```

메서드

2. Cpp

코드 구조 설명 - main.cpp

```
1 bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL$ cd 1_hello_cpp/hello_cpp/  
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/1_hello_cpp/hello_cpp$ mkdir build && cd build  
j bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/1_hello_cpp/hello_cpp/build$ cmake .. && make -j4  
-- The C compiler identification is GNU 11.4.0  
-- The CXX compiler identification is GNU 11.4.0  
e -- Detecting C compiler ABI info  
-- Detecting C compiler ABI info - done  
C -- Check for working C compiler: /usr/bin/cc - skipped  
t -- Detecting C compile features  
-- Detecting C compile features - done  
C -- Detecting CXX compiler ABI info  
-- Detecting CXX compiler ABI info - done  
-- Check for working CXX compiler: /usr/bin/c++ - skipped  
-- Detecting CXX compile features  
-- Detecting CXX compile features - done  
-- Configuring done  
-- Generating done  
-- Build files have been written to: /home/bssm/BSSM-winter-camp-PCL/1_hello_cpp/hello_cpp/build  
[ 33%] Building CXX object CMakeFiles/hello_cpp.dir/src/main.cpp.o  
[ 66%] Building CXX object CMakeFiles/hello_cpp.dir/src/person.cpp.o  
[100%] Linking CXX executable hello_cpp  
[100%] Built target hello_cpp
```

2. Cpp

코드 구조 설명 - main.cpp

```
[100%] Built target hello_cpp  
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/1_hello_cpp/hello_cpp/build$ ./hello_cpp  
이름: 홍길동, 나이: 20세  
홍길동의 취미: 축구, 독서
```

2. Cpp

코드 구조 설명 - main.cpp

Vector 는 배열 느낌?

→ [0, 1] 이런식으로

그럼 저 코드에서는

students: [르끌레르, 알론소]

```
// 5. Vector로 여러 학생 관리
std::vector<Person> students;

students.push_back(Person("르끌레르", 28));
students.push_back(Person("알론소", 44));

std::cout << "\n전체 학생 수: " << students.size() << "명" << std::endl;

for (auto& student : students) {
    student.print_info();
    student.print_hobbies();
}
```

2. Cpp

코드 구조 설명 - main.cpp

```
전체 학생 수 : 2명  
이름 : 르끌레르, 나이 : 28세  
르끌레르의 취미 :  
이름 : 알론소, 나이 : 44세  
알론소의 취미 :
```

```
huan@huan-virtual-machine: /
```

3. Ros2 cpp

3. Ros2 cpp

Ros2 란 무엇인가?

로봇 운영체제 (Robot Operating System)

로봇 분야에서 많이 사용함 그냥 거의 국룰 느낌스

3. Ros2 cpp

Ros2 란 무엇인가?

근데 하드한걸 돌리면 그닥 좋지가 않아요

**그래서 저희 회사는 Ros를 최대한 안쓰는 방향으로
가려고 하는데 일단은 배워두셔라**

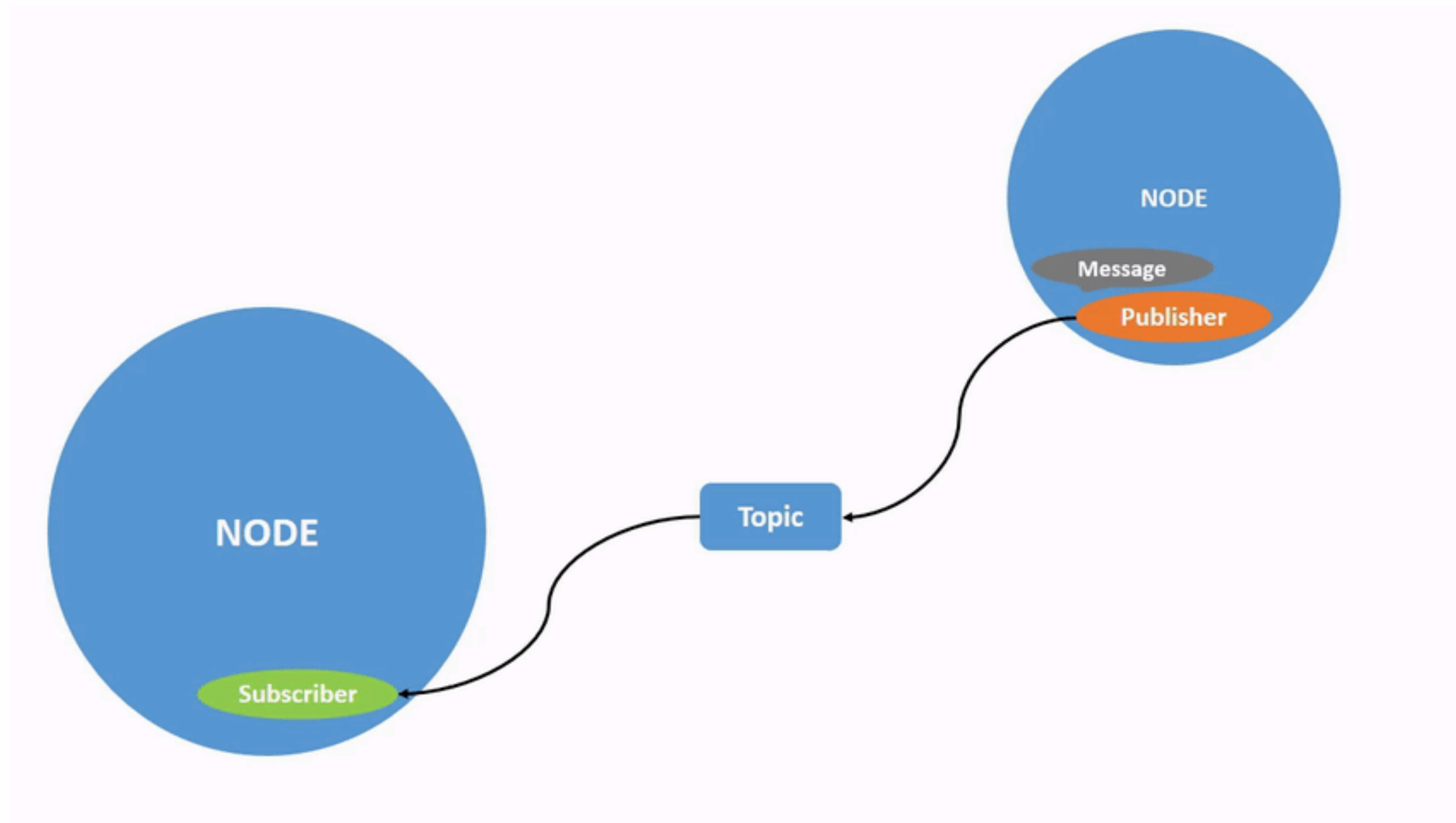
3. Ros2 cpp

Ros2 Topic

**솔직히 4시간안에 Ros 를 다 배울 수가 없어서
Ros의 가장 기초인 Topic 만 하려고 합니다.**

3. Ros2 cpp

Ros2 Topic



3. Ros2 cpp

Ros2 란 무엇인가?

Subscriber



영상올림



Publisher



업업

3. Ros2 cpp

Ros2 Topic

솔직히 처음은 좀 어려우니까 코드로 감 잡아봅시다

3. Ros2 cpp

Ros2 Topic

해당 위치에서
colcon build 커맨드 실행

▼ 2_hello_ros2

> hello_publisher

> hello_subscription

3. Ros2 cpp

Ros2 Topic



`colcon build` successful

/home/bssm/BSSM-winter-camp-PCL/2_hello_ros2

```
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ colcon build
Starting >>> hello_publisher
Starting >>> hello_subscription
Finished <<< hello_publisher [7.79s]
Finished <<< hello_subscription [10.1s]
Summary: 2 packages finished [10.4s]
```

3. Ros2 cpp

Ros2 Topic

build/

install/

log/

→ 폴더가 새로 생김

▼ 2_hello_ros2

> build

> hello_publisher

> hello_subscription

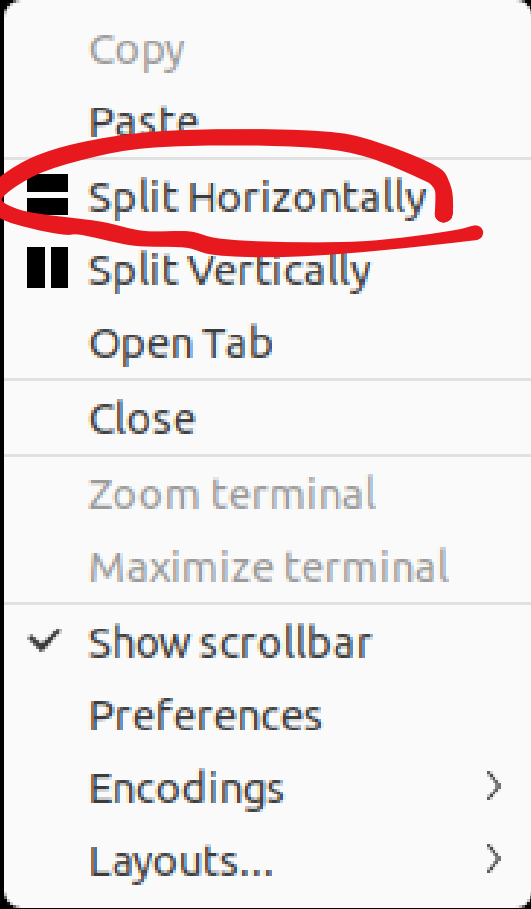
> install

> log

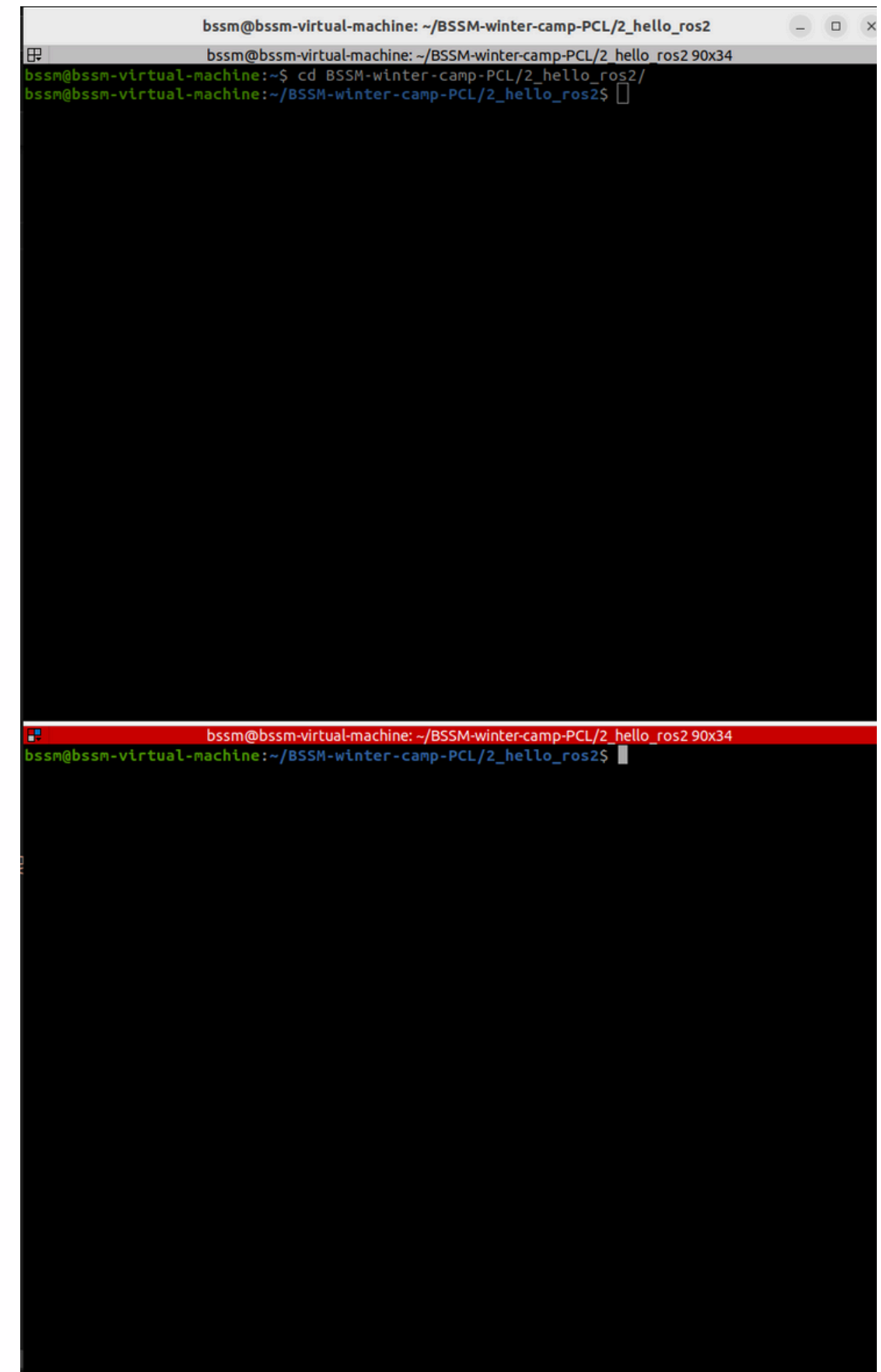
3. Ros2 cpp

Ros2 Topic

```
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/2_hello_ros2
bssm@bssm-virtual-machine:~$ cd BSSM-winter-camp-PCL/2_hello_ros2/
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$
```



A context menu is displayed over the terminal window. The menu items are: Copy, Paste, Split Horizontally (highlighted with a red circle), Split Vertically, Open Tab, Close, Zoom terminal, Maximize terminal, Show scrollbar (checked), Preferences, Encodings, and Layouts... The 'Split Horizontally' option is the third item in the menu, preceded by a square icon with two horizontal lines.



3. Ros2 cpp

Ros2 Topic

**colcon 을 사용하였다면 무조건 다음 명령어를
실행해야합니다. (각 터미널에!!!!)**

source install/setup.bash

3. Ros2 cpp

Ros2 Topic

**그 후 각 터미널에 다음 명령어를 입력하여 코드를
실행해주세요.**

- 1. `ros2 run hello_publisher talker`**
- 2. `ros2 run hello_subscription listener`**

3. Ros2 cpp

Ros2 Topic

```
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/2_hello_ros2
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/2_hello_ros2 90x14
bssm@bssm-virtual-machine:~$ cd BSSM-winter-camp-PCL/2_hello_ros2/
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ source install/setup.bash
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 node list
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 run hello_publihser ta
lker
Package 'hello_publihser' not found
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 run hello_publisher ta
lker
[INFO] [1766673796.795615950] [talker]: publish: 0
[INFO] [1766673797.795637124] [talker]: publish: 1
[INFO] [1766673798.795561328] [talker]: publish: 2
[INFO] [1766673799.795616478] [talker]: publish: 3
[INFO] [1766673800.795583223] [talker]: publish: 4
[INFO] [1766673801.795625029] [talker]: publish: 5
[INFO] [1766673802.795598259] [talker]: publish: 6
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/2_hello_ros2 90x54
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ source install/setup.bash
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 run hello_subscription
listener
[INFO] [1766673806.259088907] [listener]: Subscribing to topic: counter
[INFO] [1766673806.796006700] [listener]: I heard: 10
[INFO] [1766673807.795899889] [listener]: I heard: 11
[INFO] [1766673808.795889991] [listener]: I heard: 12
[INFO] [1766673809.795861549] [listener]: I heard: 13
[INFO] [1766673810.795811432] [listener]: I heard: 14
[INFO] [1766673811.795980186] [listener]: I heard: 15
[INFO] [1766673812.795859421] [listener]: I heard: 16
[INFO] [1766673813.795980076] [listener]: I heard: 17
[INFO] [1766673814.795904449] [listener]: I heard: 18
[INFO] [1766673815.795914040] [listener]: I heard: 19
[INFO] [1766673816.795941322] [listener]: I heard: 20
```

3. Ros2 cpp

Ros2 Topic

**이제 어떻게 동작하는지 확인했으니 코드를
확인합시다!**

3. Ros2 cpp

Ros2 Topic - Publisher

폴더 구조는 다음과 같습니다.

```
▼ 2_hello_ros2
  > build
  ▼ hello_publisher
    ▼ include
      ➤ simple_publisher.hpp
    ▼ src
      ➤ main.cpp
      ➤ simple_publisher.cpp
```

3. Ros2 cpp

Ros2 Topic - Publisher main.cpp

딱히 별게 없어요

rclcpp::spin == while (true)

```
1  #include "simple_publisher.hpp"
2
3  #include <memory>
4  #include "rclcpp/rclcpp.hpp"
5
6  int main(int argc, char ** argv)
7  {
8      rclcpp::init(argc, argv);
9      rclcpp::spin(std::make_shared<Talker>());
10     rclcpp::shutdown();
11     return 0;
12 }
13
```

3. Ros2 cpp

Ros2 Topic - Publisher publisher

메시지 타입은 무조건 같아야함!!

```
simple_publisher.hpp × ... person.hpp person.cpp × simple_publisher.cpp × main.cpp
2_hello_ros2 > hello_publisher > include > simple_publisher.hpp
1  #pragma once
2
3  #include <chrono>
4  #include <memory>
5
6  #include "rclcpp/rclcpp.hpp"
7  #include "std_msgs/msg/int32.hpp"
8
9  using namespace std::chrono_literals;
10
11 class Talker : public rclcpp::Node
12 {
13 public:
14     Talker();
15
16 private:
17     rclcpp::Publisher<std_msgs::msg::Int32>::SharedPtr pub_;
18     rclcpp::TimerBase::SharedPtr timer_;
19     int32_t count_;
20 };
21
22
person.cpp
2_hello_ros2 > hello_publisher > ~/BSSM-winter-camp-PCL/1_hello_cpp/hello_cpp/src/person.cpp
1  #include "simple_publisher.hpp"
2
3  Talker::Talker()
4      : Node("talker"), count_(0)
5  {
6      // QoS 설정: Reliable, Keep Last 10
7      auto qos = rclcpp::QoS(rclcpp::KeepLast(10))
8          .reliable()
9          .durability_volatile();
10
11     pub_ = create_publisher<std_msgs::msg::Int32>("counter", qos);
12
13     timer_ = create_wall_timer(
14         1s, [this]()
15         {
16             std_msgs::msg::Int32 msg;
17             msg.data = count_++;
18
19             RCLCPP_INFO(get_logger(), "publish: %d", msg.data);
20             pub_>publish(msg);
21         });
22 }
```

3. Ros2 cpp

Ros2 Topic - Subscription

counter topic 을 구독하여
값을 계속 출력

```
person.hpp  person.cpp  simple_publisher.cpp  simple_subscriber.hpp X
2_hello_ros2 > hello_subscription > include > simple_subscriber.hpp
1  #pragma once
2
3  #include <memory>
4
5  #include "rclcpp/rclcpp.hpp"
6  #include "std_msgs/msg/int32.hpp"
7
8  class Listener : public rclcpp::Node
9  {
10 public:
11     Listener();
12
13 private:
14     void topic_callback(const std_msgs::msg::Int32::ConstSharedPtr msg) const;
15     rclcpp::Subscription<std_msgs::msg::Int32>::SharedPtr sub_;
16 };
17
simple_subscriber.cpp X
2_hello_ros2 > hello_subscription > src > simple_subscriber.cpp
3  Listener::Listener()
4  : Node("listener")
5  {
6      // QoS 설정: Reliable, Keep Last 10 (Publisher와 동일하게 설정)
7      auto qos = rclcpp::QoS(rclcpp::KeepLast(10))
8          .reliable()
9          .durability_volatile();
10
11      sub_ = create_subscription<std_msgs::msg::Int32>(
12          "counter", qos, std::bind(&Listener::topic_callback, this, std::placeholders::_1));
13      RCLCPP_INFO(this->get_logger(), "Subscribing to topic: counter");
14  }
15
16 void Listener::topic_callback(const std_msgs::msg::Int32::ConstSharedPtr msg) const
17 {
18     RCLCPP_INFO(this->get_logger(), "I heard: %d", msg->data);
19 }
20
```

3. Ros2 cpp

Ros2 Topic

```
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/2_hello_ros2
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/2_hello_ros2 90x14
bssm@bssm-virtual-machine:~$ cd BSSM-winter-camp-PCL/2_hello_ros2/
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ source install/setup.bash
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 node list
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 run hello_publihser ta
lker
Package 'hello_publihser' not found
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 run hello_publisher ta
lker
[INFO] [1766673796.795615950] [talker]: publish: 0
[INFO] [1766673797.795637124] [talker]: publish: 1
[INFO] [1766673798.795561328] [talker]: publish: 2
[INFO] [1766673799.795616478] [talker]: publish: 3
[INFO] [1766673800.795583223] [talker]: publish: 4
[INFO] [1766673801.795625029] [talker]: publish: 5
[INFO] [1766673802.795598259] [talker]: publish: 6
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2 90x54
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ source install/setup.bash
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 run hello_subscription
listener
[INFO] [1766673806.259088907] [listener]: Subscribing to topic: counter
[INFO] [1766673806.796006700] [listener]: I heard: 10
[INFO] [1766673807.795899889] [listener]: I heard: 11
[INFO] [1766673808.795889991] [listener]: I heard: 12
[INFO] [1766673809.795861549] [listener]: I heard: 13
[INFO] [1766673810.795811432] [listener]: I heard: 14
[INFO] [1766673811.795980186] [listener]: I heard: 15
[INFO] [1766673812.795859421] [listener]: I heard: 16
[INFO] [1766673813.795980076] [listener]: I heard: 17
[INFO] [1766673814.795904449] [listener]: I heard: 18
[INFO] [1766673815.795914040] [listener]: I heard: 19
[INFO] [1766673816.795941322] [listener]: I heard: 20
```

3. Ros2 cpp

Ros2 Topic

```
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/2_hello_ros2
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/2_hello_ros2 90x14
bssm@bssm-virtual-machine:~$ cd BSSM-winter-camp-PCL/2_hello_ros2/
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ source install/setup.bash
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 node list
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 run hello_publihser ta
lker
Package 'hello_publihser' not found
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 run hello_publisher ta
lker
[INFO] [1766673796.795615950] [talker]: publish: 0
[INFO] [1766673797.795637124] [talker]: publish: 1
[INFO] [1766673798.795561328] [talker]: publish: 2
[INFO] [1766673799.795616478] [talker]: publish: 3
[INFO] [1766673800.795583223] [talker]: publish: 4
[INFO] [1766673801.795625029] [talker]: publish: 5
[INFO] [1766673802.795598259] [talker]: publish: 6
bssm@bssm-virtual-machine: ~/BSSM-winter-camp-PCL/2_hello_ros2 90x54
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ source install/setup.bash
bssm@bssm-virtual-machine:~/BSSM-winter-camp-PCL/2_hello_ros2$ ros2 run hello_subscription
listener
[INFO] [1766673806.259088907] [listener]: Subscribing to topic: counter
[INFO] [1766673806.796006700] [listener]: I heard: 10
[INFO] [1766673807.795899889] [listener]: I heard: 11
[INFO] [1766673808.795889991] [listener]: I heard: 12
[INFO] [1766673809.795861549] [listener]: I heard: 13
[INFO] [1766673810.795811432] [listener]: I heard: 14
[INFO] [1766673811.795980186] [listener]: I heard: 15
[INFO] [1766673812.795859421] [listener]: I heard: 16
[INFO] [1766673813.795980076] [listener]: I heard: 17
[INFO] [1766673814.795904449] [listener]: I heard: 18
[INFO] [1766673815.795914040] [listener]: I heard: 19
[INFO] [1766673816.795941322] [listener]: I heard: 20
```

4. Pcl 데이터 찍먹

4. Pcl 데이터 찍먹

Pcl 데이터 핸들링

해당 부분부터는 서론이 조금 들어갈거예요

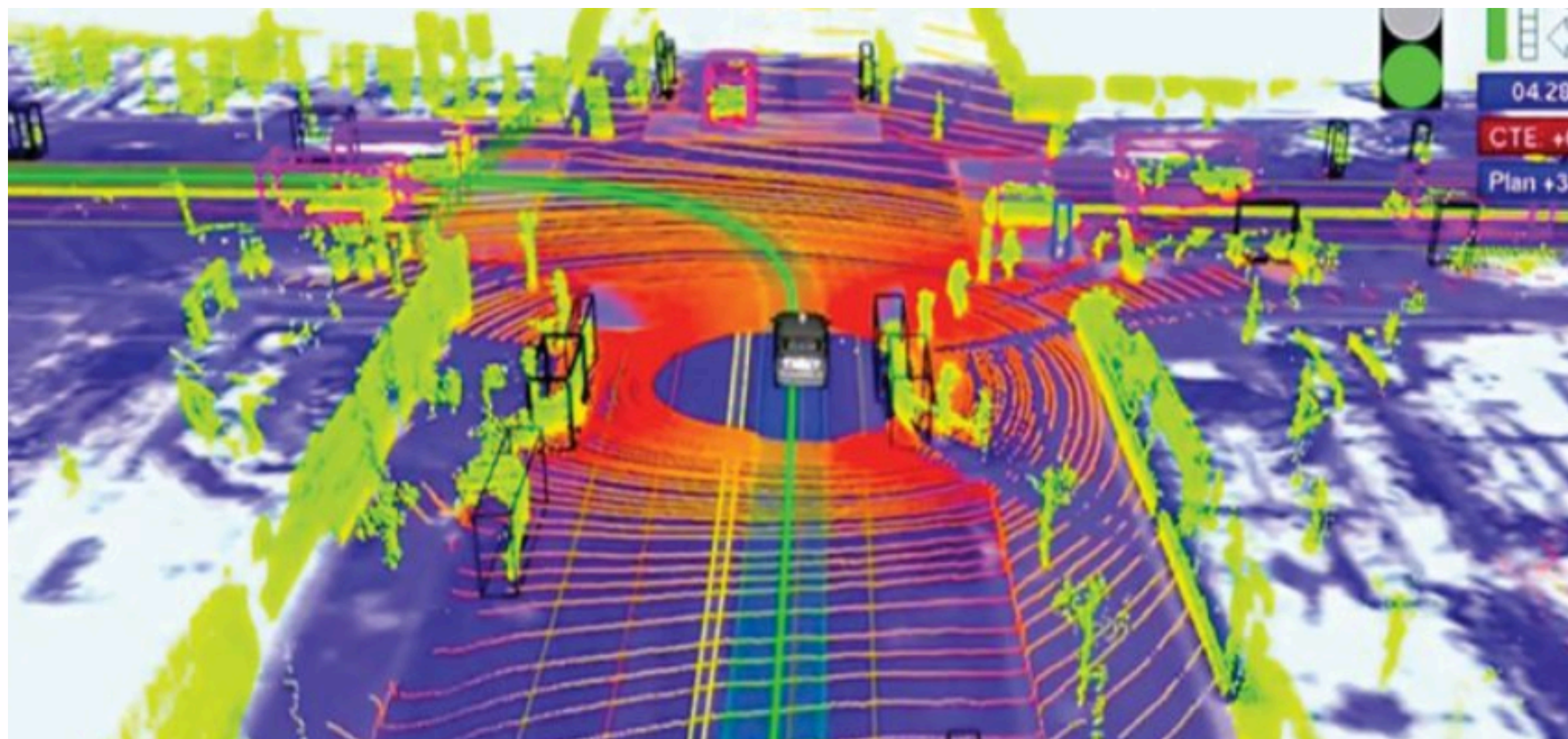
4. Pcl 데이터 짹먹

Pcl 데이터 핸들링

제 회사에서는 어떤 식으로 Ply 데이터를 사용하는지
보여드릴건데 저도 잘 못해요

4. Pcl 데이터 찍먹

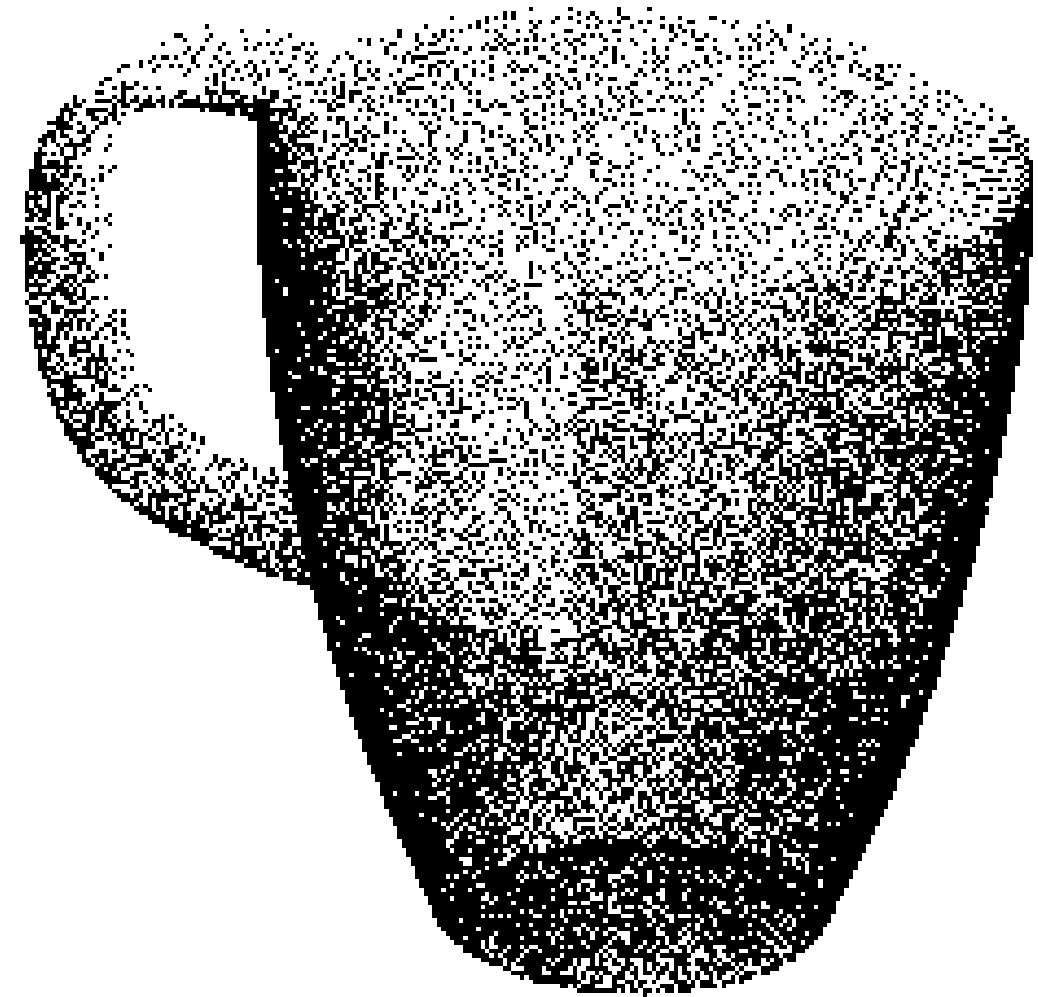
Pcl 데이터 핸들링



4. Pcl 데이터 찍먹

Pcl 데이터 핸들링

Pointcloud



4. Pcl 데이터 짹먹

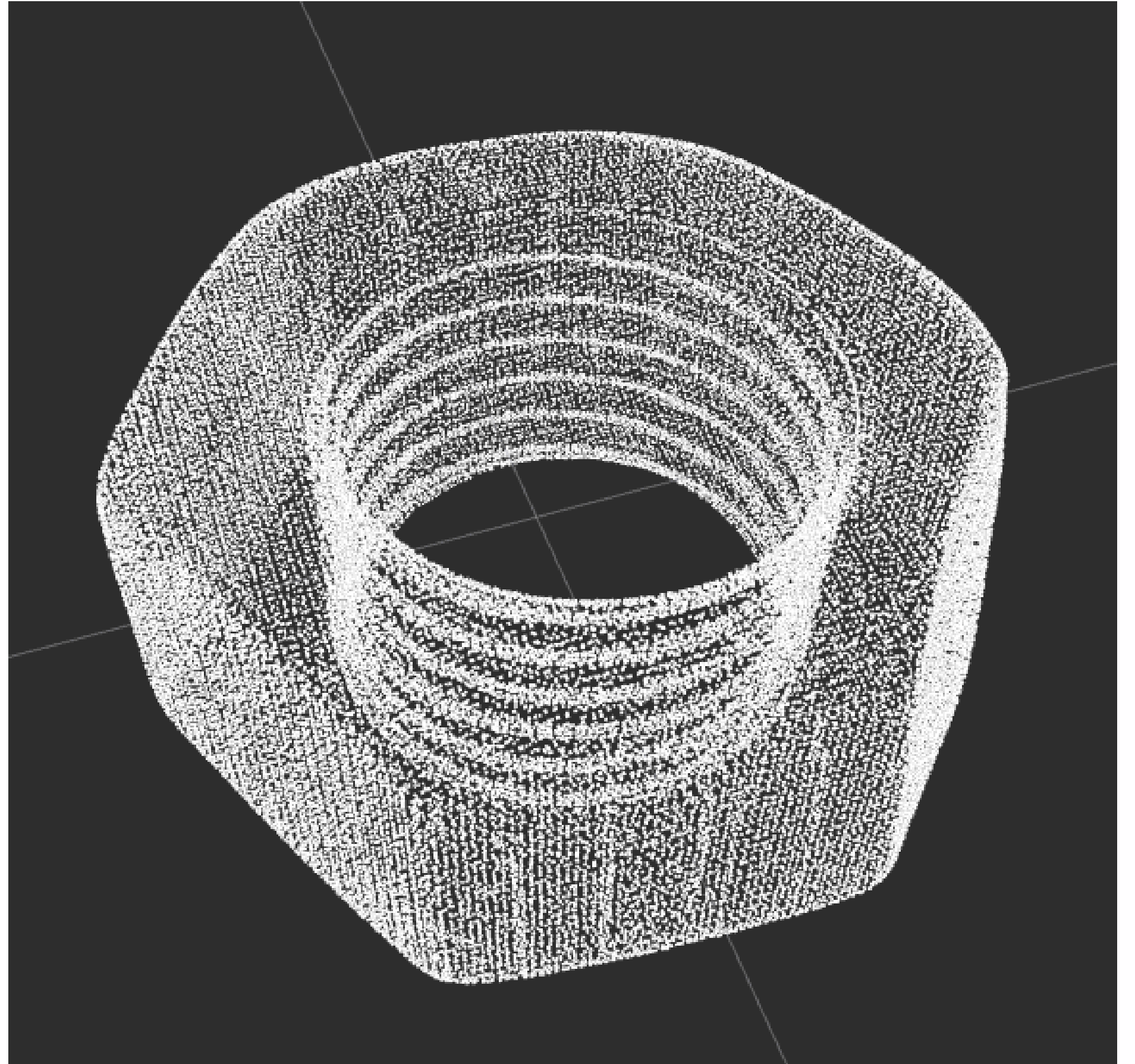
Pcl 데이터 핸들링

제품의 중앙 홀을 찾는 알고리즘 실습

4. Ply 데이터 찍먹

Ply 데이터 핸들링

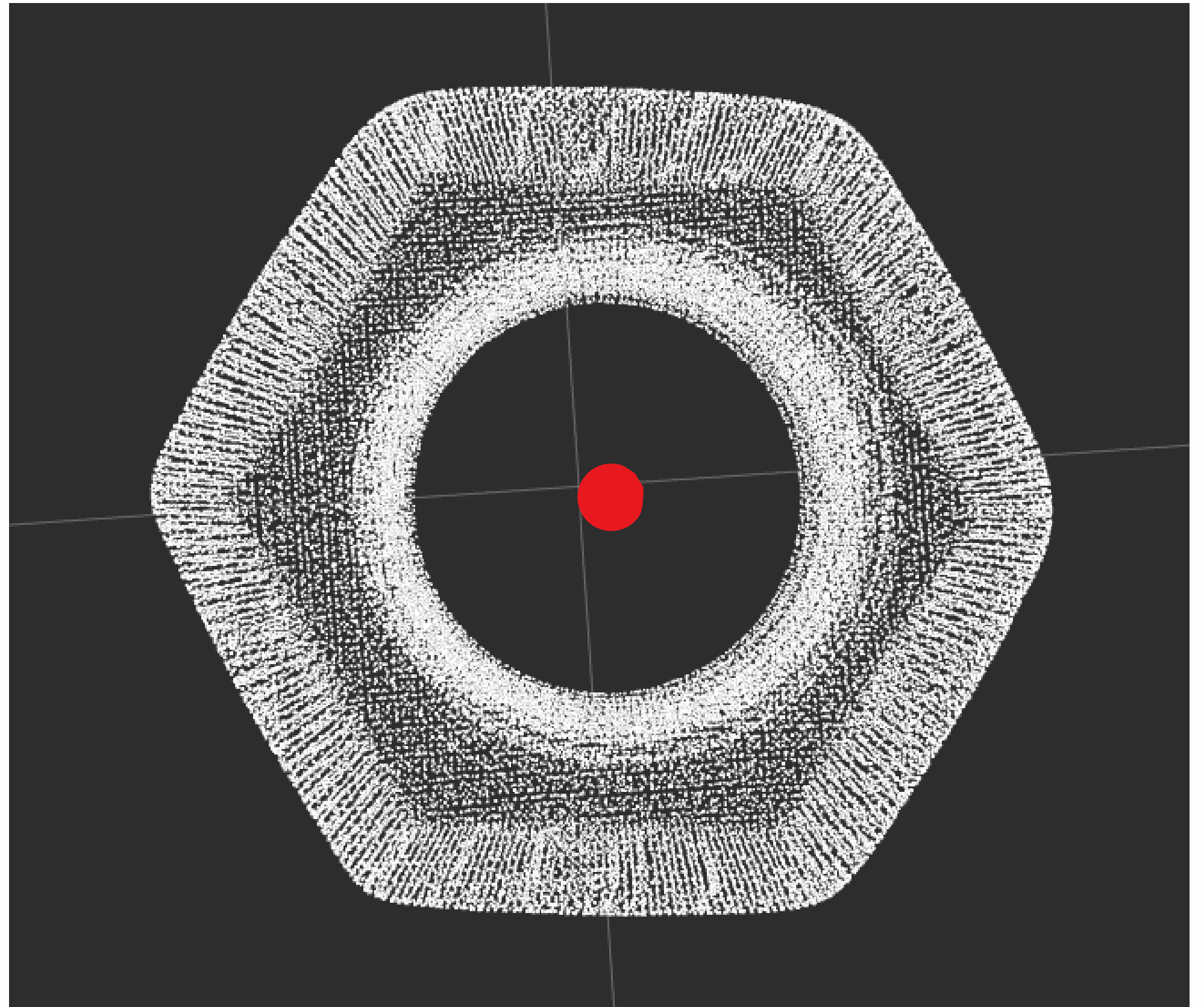
오늘 사용할 데이터



4. Ply 데이터 찍먹

Ply 데이터 핸들링

빨간색 점의 위치를 찾는게
목표!



4. Ply 데이터 찍먹

Ply 데이터 핸들링

폴더 구조는 다음과 같습니다.

```
▼ 3_hello_ply
  > build
  > data
  ▼ hello_ply
    > config
    ▼ include
      ↗ ply_helper.hpp
    ▼ src
      ↗ main.cpp
      ↗ ply_helper.cpp
  M CMakeLists.txt
  📄 package.xml
  > install
  > log
```

4. Ply 데이터 찍먹

Ply 데이터 핸들링

현재 코드가 약 600 줄 정도?

좀 많이 긴데 이걸 실습 하면서 설명해드릴게요

실행 코드는 ply_helper.cpp 안에 process() 함수

4. Ply 데이터 찍먹

Ply 데이터 핸들링

주석에 어떤 흐름으로 가는지
적어놨어요

```
void PlyHelper::process() {  
    /*  
    전체 흐름 설명  
  
    1. 포인트 클라우드 로드  
    2. 포인트 클라우드 중심 이동 및 스케일링  
    3. 포인트 클라우드 다운샘플링  
    4. 다운샘플링된 포인트 클라우드에서 Z 축이 0 이상인 포인트만 필터링 ( 노란색 )  
    5. RANSAC으로 위쪽 평면 찾기 ( 초록색 )  
    6. 외곽선 추출 (평면 포인트에서 추출) ( 빨간색 )  
    7. 찾은 외곽 포인트들 중 가장 작은 포인트 클라우드를 중심 포인트 클라우드로 설정 ( 파란색 )  
    8. 중앙 포인트 클라우드를 기준으로 사각형 생성 ( 흰색 )  
    */  
  
    // 1. 포인트 클라우드 로드  
    loadPlyFile(file_path_);  
  
    // 2. 포인트 클라우드 중심 이동 및 스케일링  
    center_and_scale_pointcloud();  
  
    // 3. 포인트 클라우드 다운샘플링  
    CloudPtr downsampled_cloud;  
    downsampling(downsampled_cloud);  
    change_pointcloud_color(downsampled_cloud, 255, 255, 255);  
    publishPointCloud(downsampled_cloud, downsampled_publisher_);  
}
```

4. Ply 데이터 찍먹

Ply 데이터 핸들링

ros2 run hello_ply ply_publisher

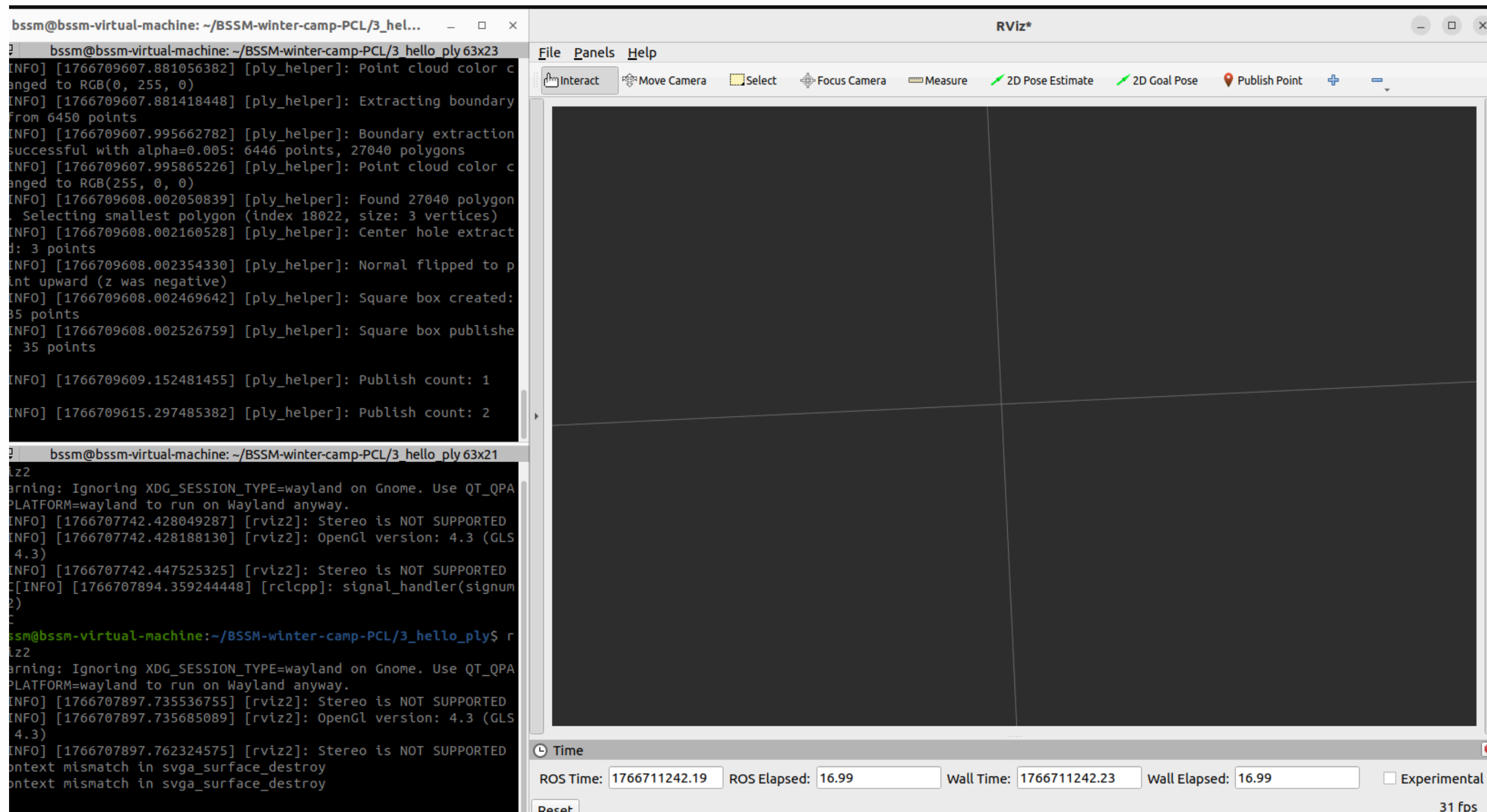
터미널 2개 실행해주세요!

colcon build 후 source 까지 했다고 가정



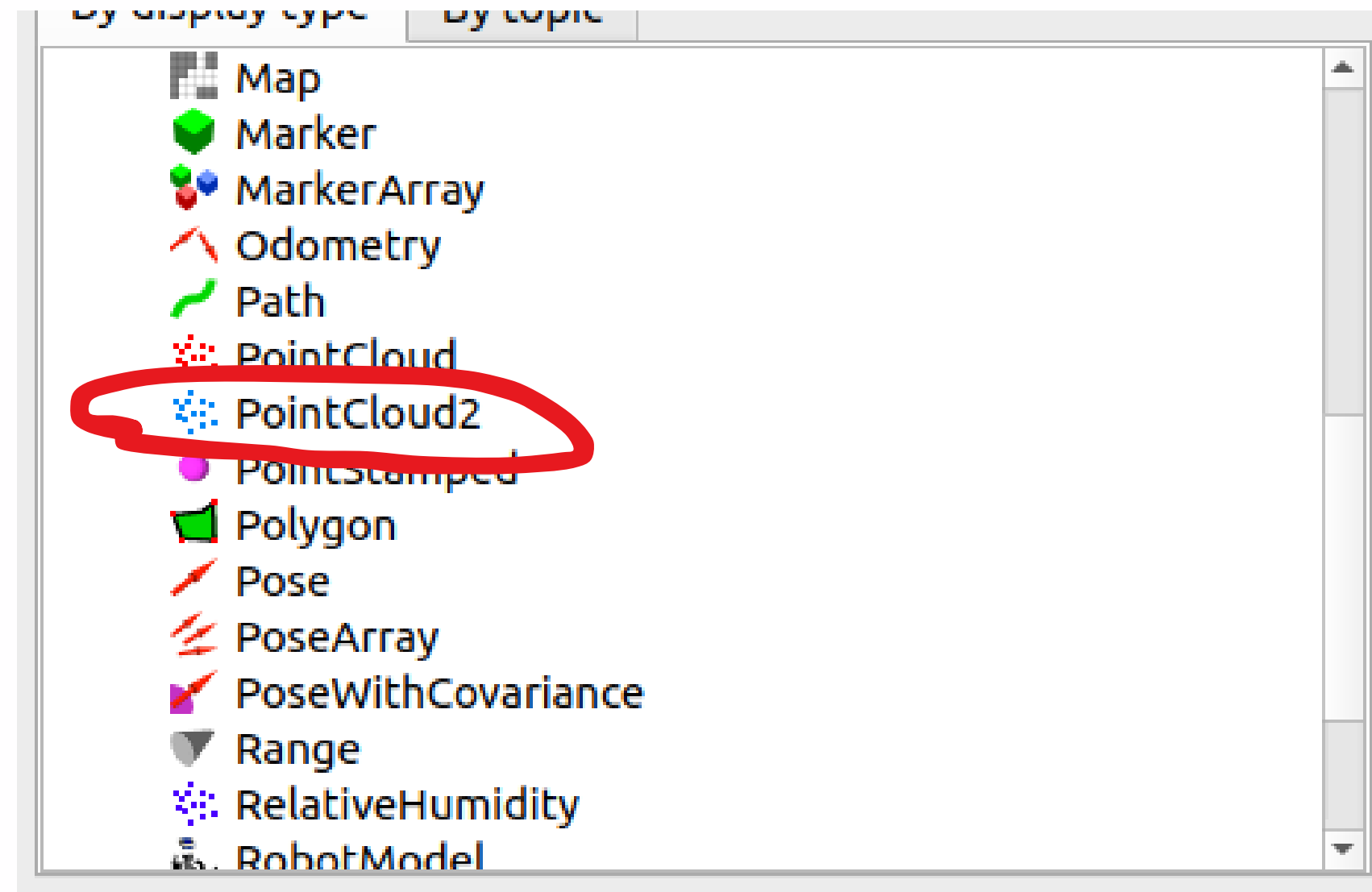
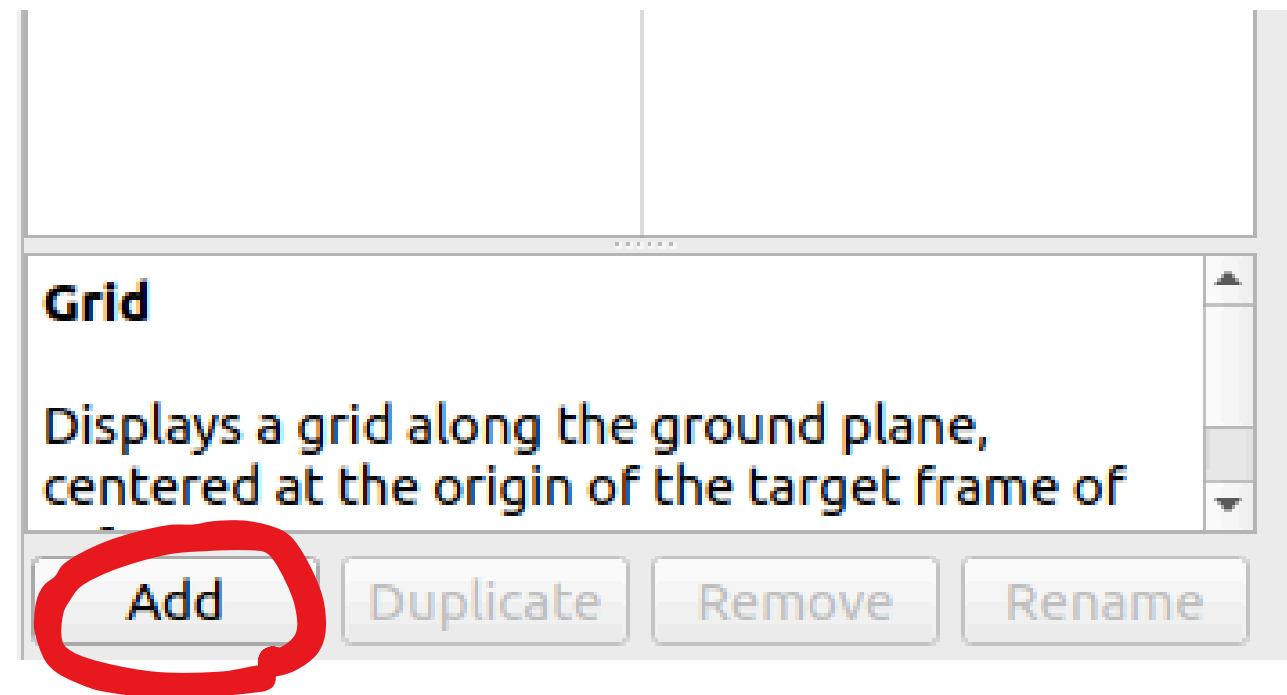
4. Ply 데이터 찍먹

Ply 데이터 핸들링



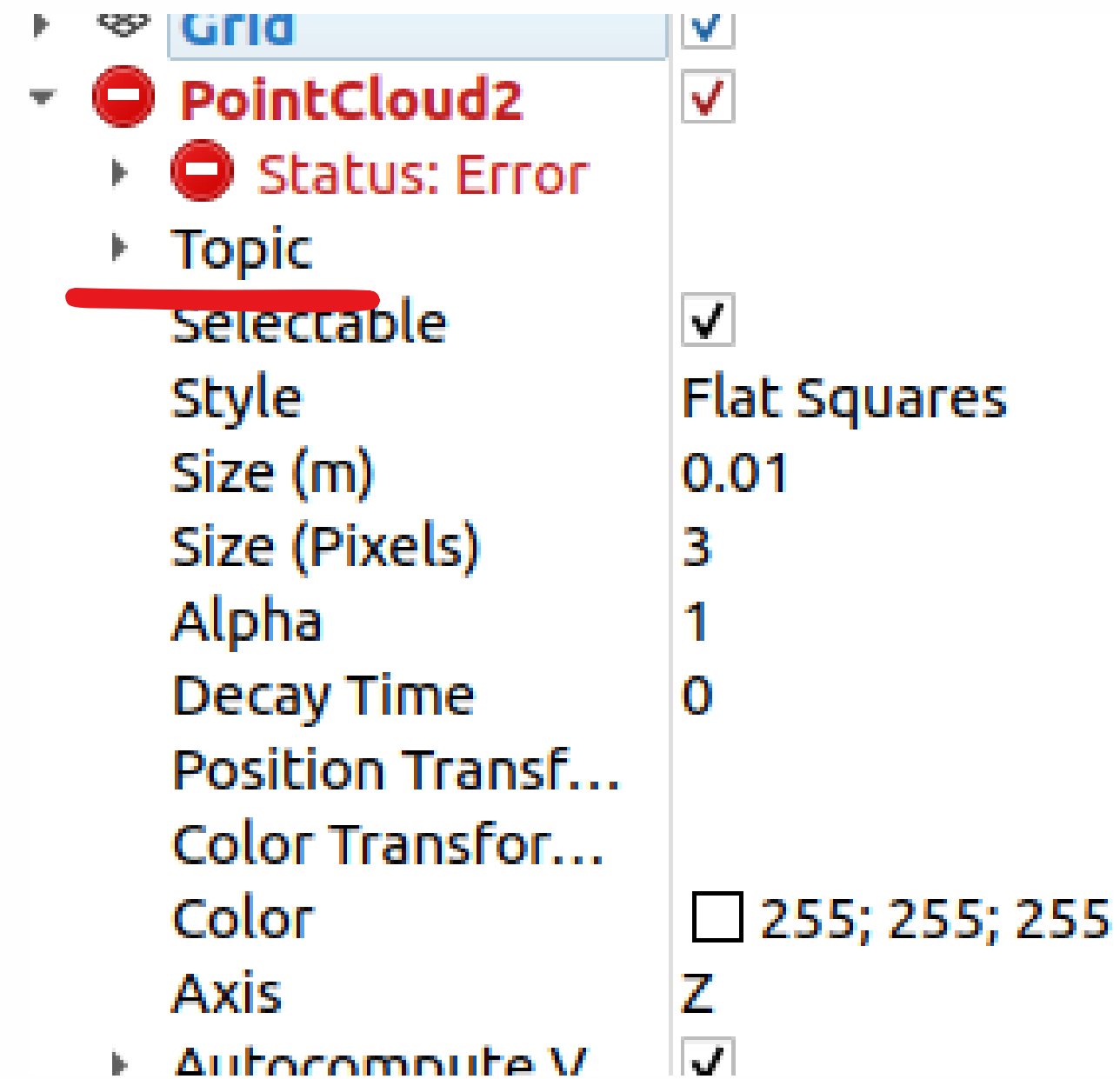
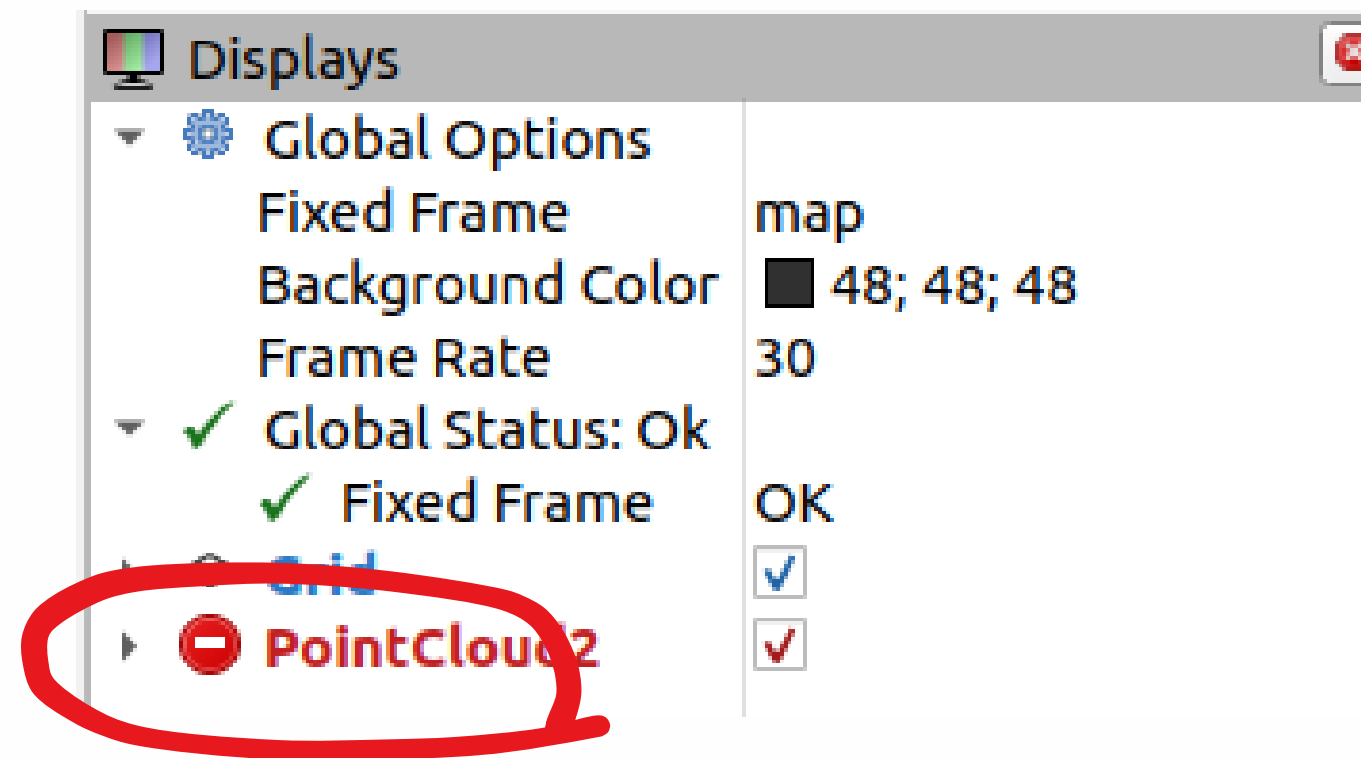
4. Ply 데이터 찍먹

Ply 데이터 핸들링



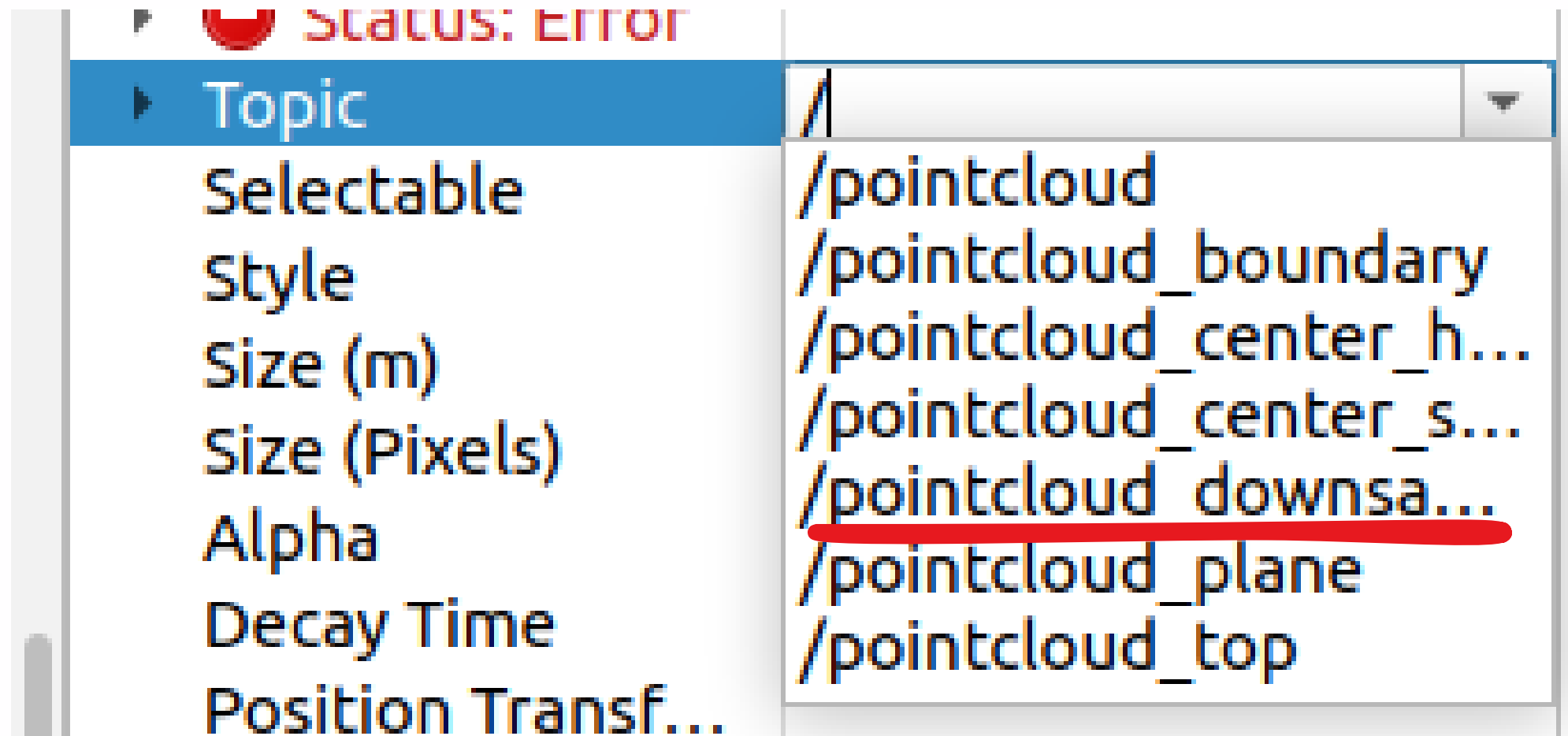
4. Ply 데이터 찍먹

Ply 데이터 핸들링



4. Ply 데이터 찍먹

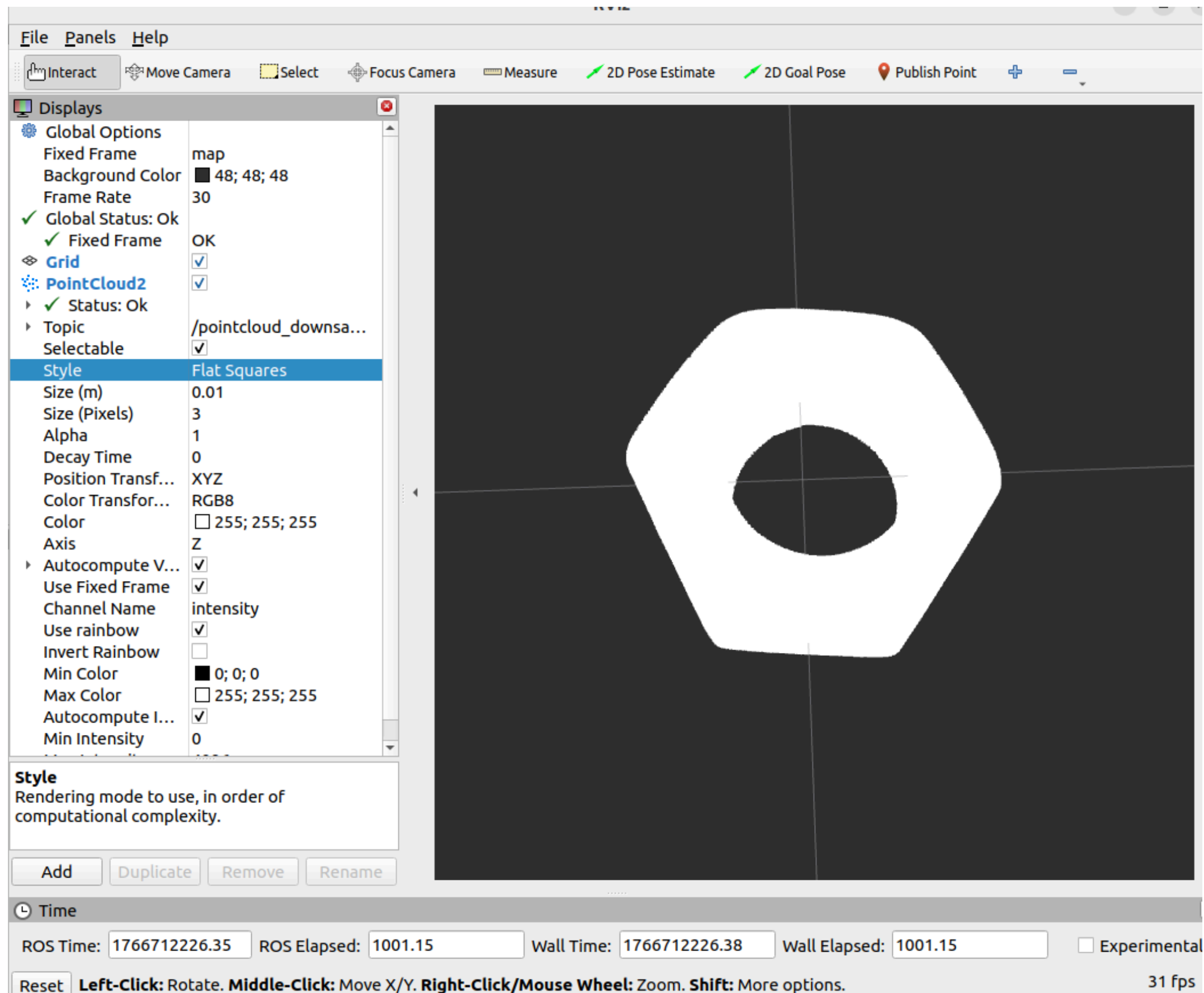
Ply 데이터 핸들링



▶ ✓ Status: Ok

4. Ply 데이터 찍먹

Ply 데이터 핸들링



4. Ply 데이터 찍먹

Ply 데이터 핸들링

Size (m) 의 값을 다음과 같이
수정해주세요

PointCloud2	☒
▶ ✓ Status: Ok	
▶ Topic	/pointcloud_downsa...
Selectable	☒
Style	Flat Squares
Size (m)	0.001
Size (Pixels)	3
Alpha	1
Decay Time	0
Position Transf...	XYZ
Color Transfor...	RGB8
Color	☐ 255; 255; 255
Axis	Z

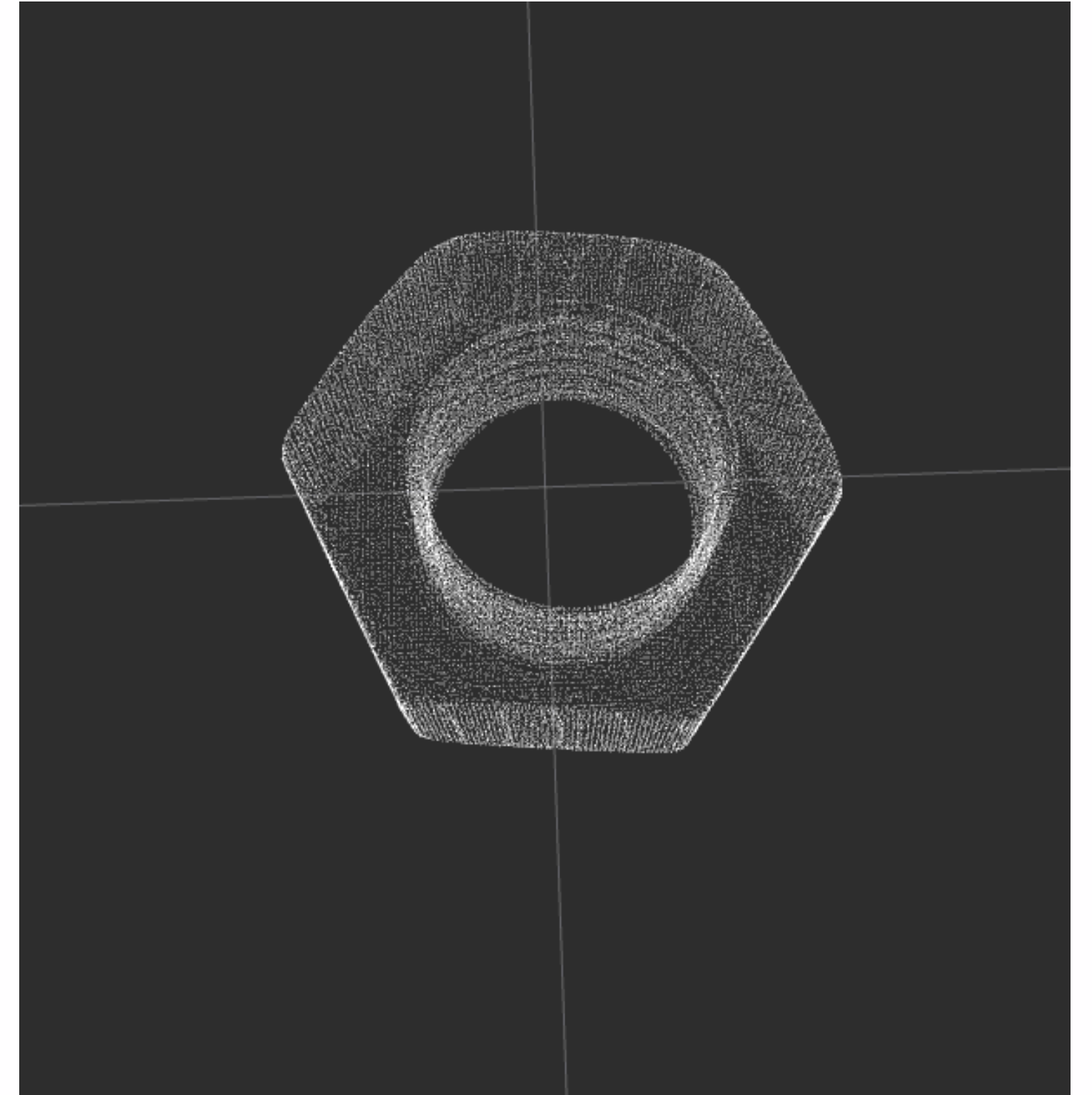
4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 1 ~ 3

```
// 1. 포인트 클라우드 로드
loadPlyFile(file_path_);

// 2. 포인트 클라우드 중심 이동 및 스케일링
center_and_scale_pointcloud();

// 3. 포인트 클라우드 다운샘플링
CloudPtr downsampled_cloud;
downsampling(downsampled_cloud);
change_pointcloud_color(downsampled_cloud, 255, 255, 255);
publishPointCloud(downsampled_cloud, downsampled_publisher_);
```



4. Ply 데이터 찍먹

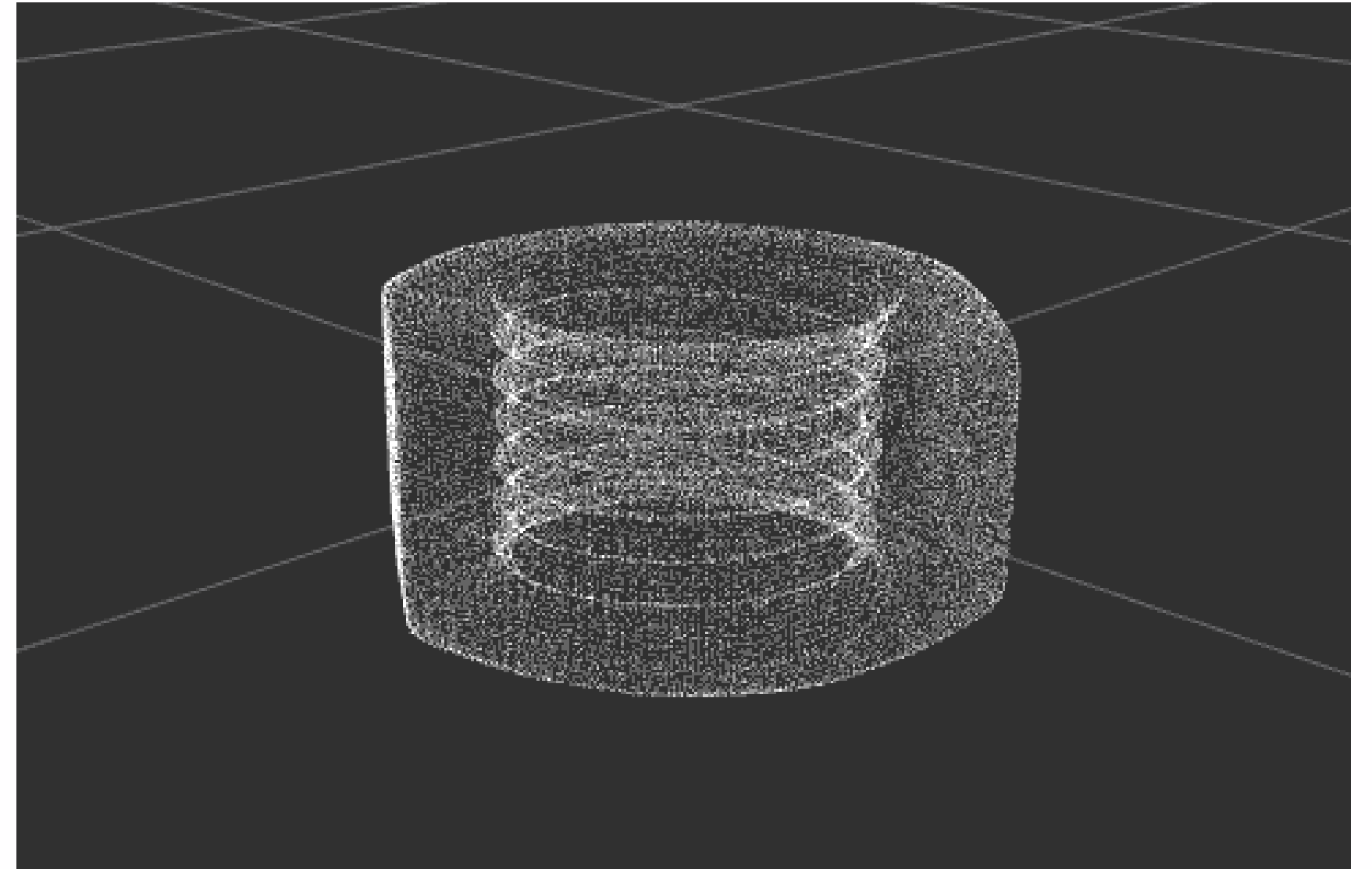
Ply 데이터 핸들링 - Step 4

```
// 4. 다운샘플링된 포인트 클라우드에서 Z 축이 0 이상인 포인트만 필터링 ( 노란색 )  
CloudPtr offset_z_cloud;  
extract_offset_z(downsampled_cloud, offset_z_cloud, 0.0f);  
change_pointcloud_color(offset_z_cloud, 255, 255, 0);  
publishPointCloud(offset_z_cloud, offset_z_cloud_publisher_);
```

4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 4

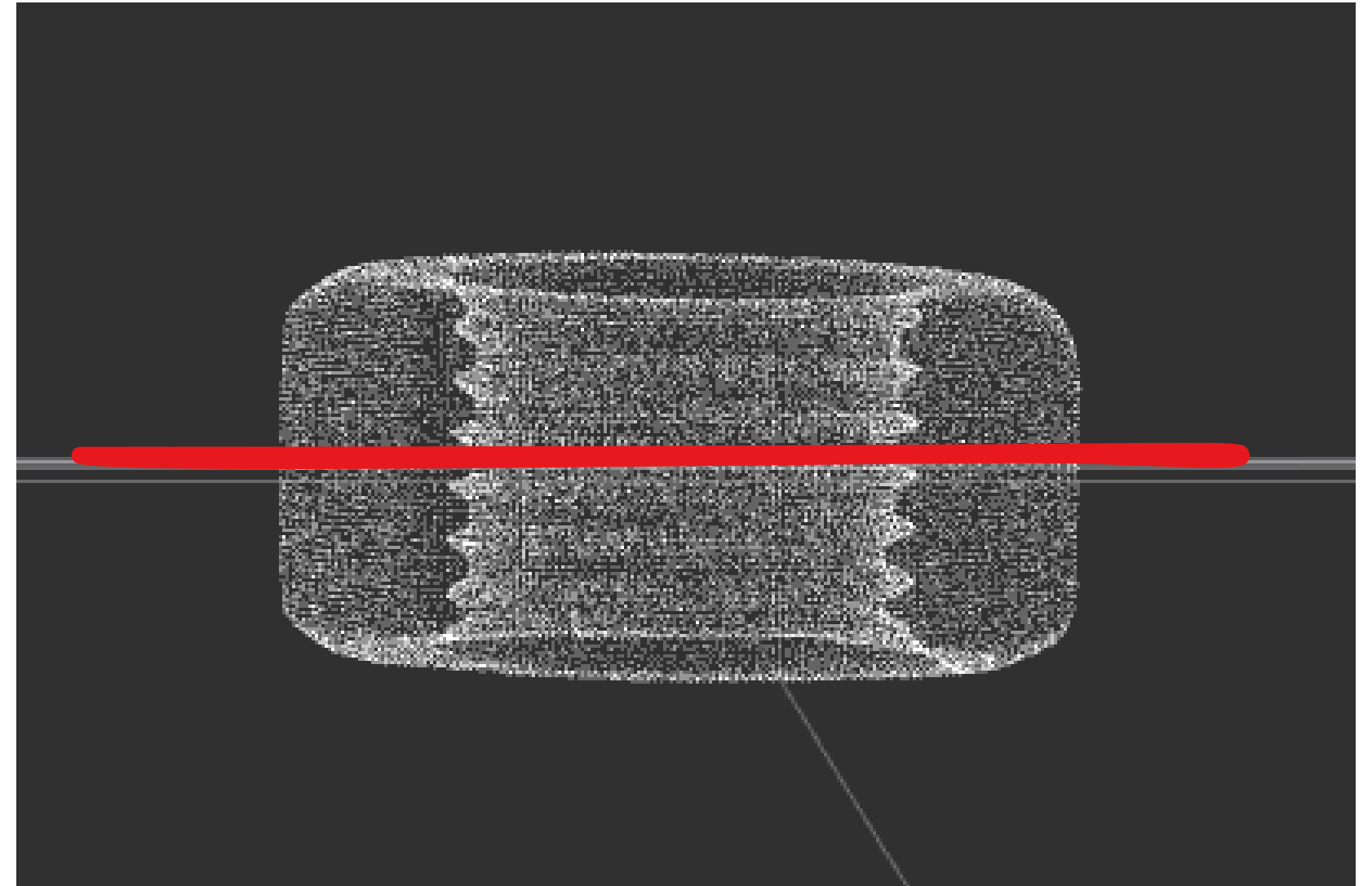
Rviz에서 흰색 선 부분이 z가 0인 부분



4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 4

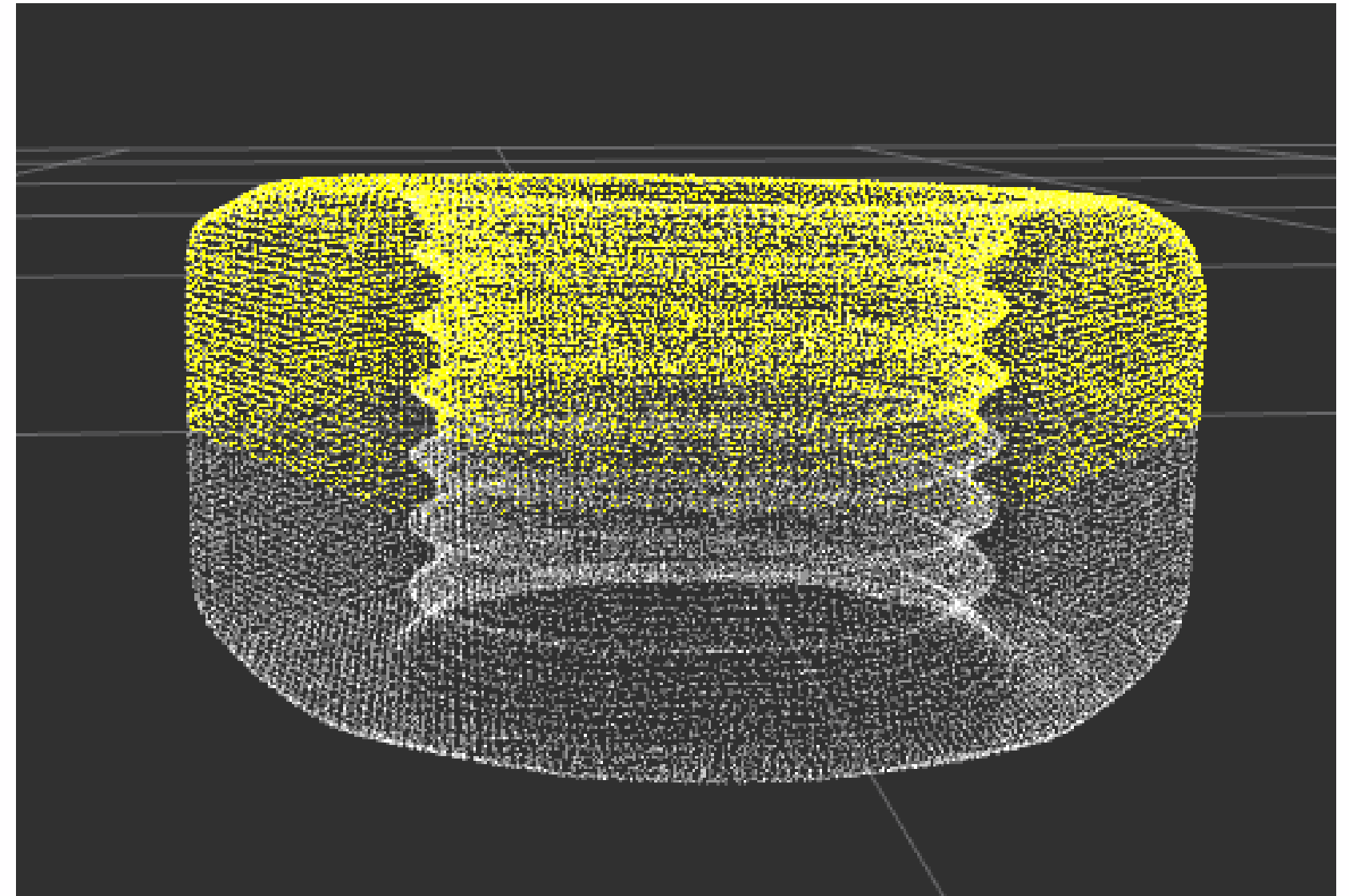
빨간색 선 위쪽이 Z가 +인 부분
즉, 위쪽만 잘라지겠죠?



4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 4

 PointCloud2	☒
▶  Status: Ok	
▶ Topic	/pointcloud_top
Selectable	☒
Style	Points
<u>Size (m)</u>	0.01
<u>Size (Pixels)</u>	1
Alpha	1
Decay Time	0
Position Transf...	XYZ
Color Transfor...	RGB8
Color	☐ 255; 255; 255
Axis	Z



4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 5

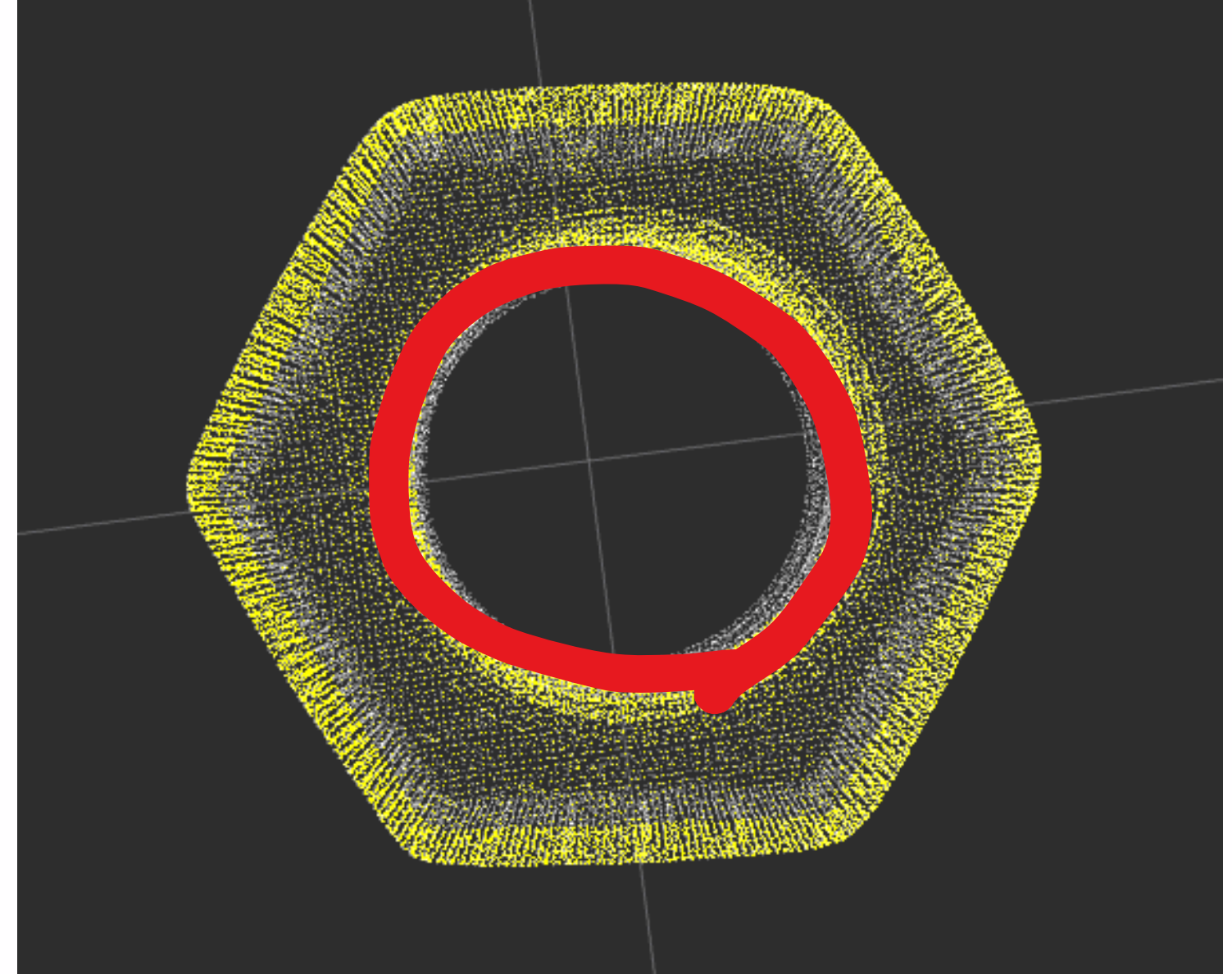
```
// 5. RANSAC으로 위쪽 평면 찾은 후 초록색으로 변경
CloudPtr plane_cloud;
pcl::ModelCoefficients::Ptr plane_coefficients;
find_top_plane_ransac(offset_z_cloud, plane_cloud, plane_coefficients);
change_pointcloud_color(plane_cloud, 0, 255, 0);
publishPointCloud(plane_cloud, plane_publisher_);
```

4. Ply 데이터 짹먹

Ply 데이터 핸들링 - Step 5

위쪽에 평면을 찾는 이유는 나중에
홀을 찾기 위함?

찾아야 할 부분이 빨간색 부분인데
그냥 찾아버리면 알고리즘에서 값이 이상함
ㅠㅠ

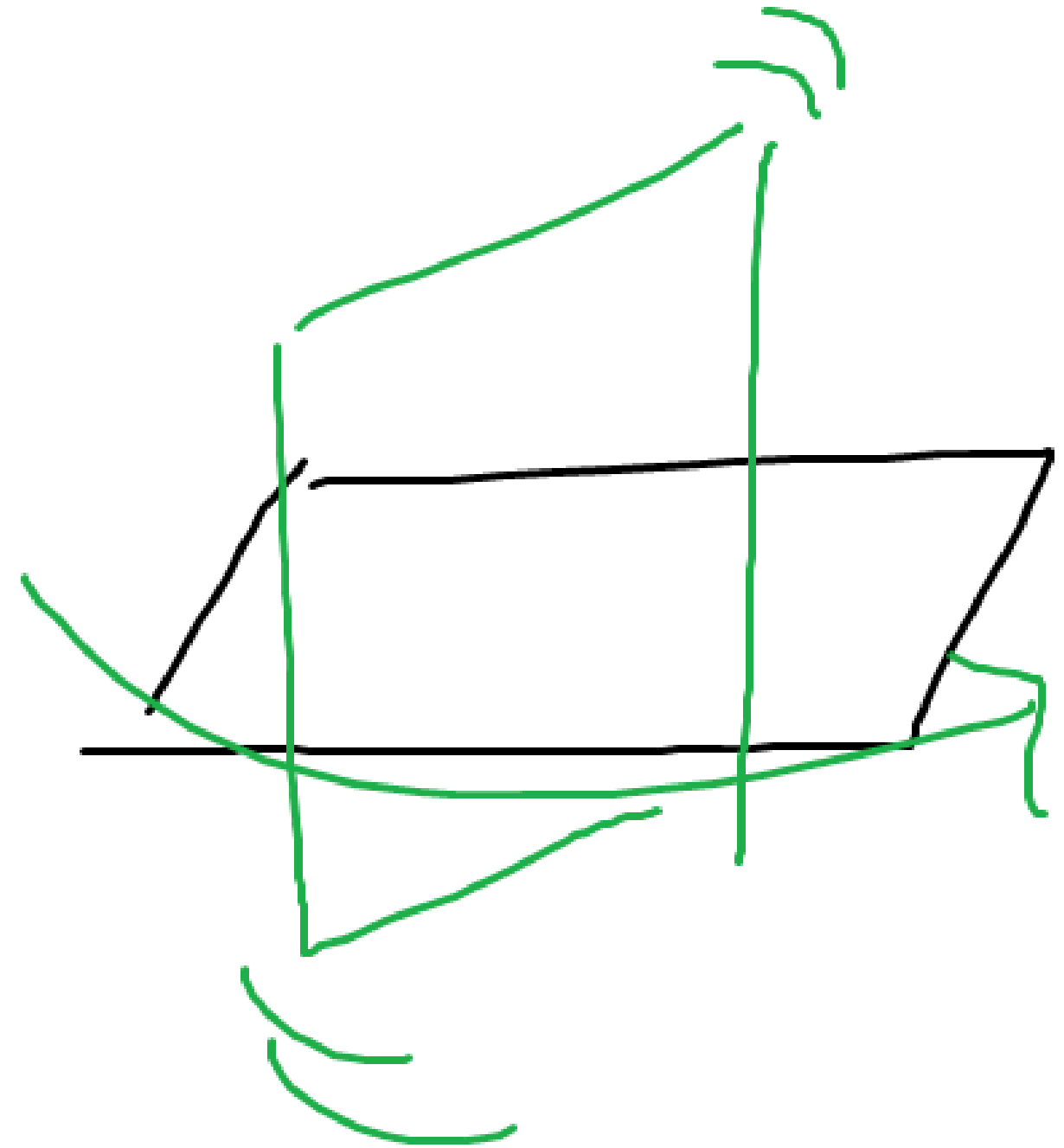


4. Ply 데이터 짝먹

Ply 데이터 핸들링 - Step 5 (RANSAC)

검은색 판을 찾고 싶다고 할 때 초록색 판이 있다고 가정하면

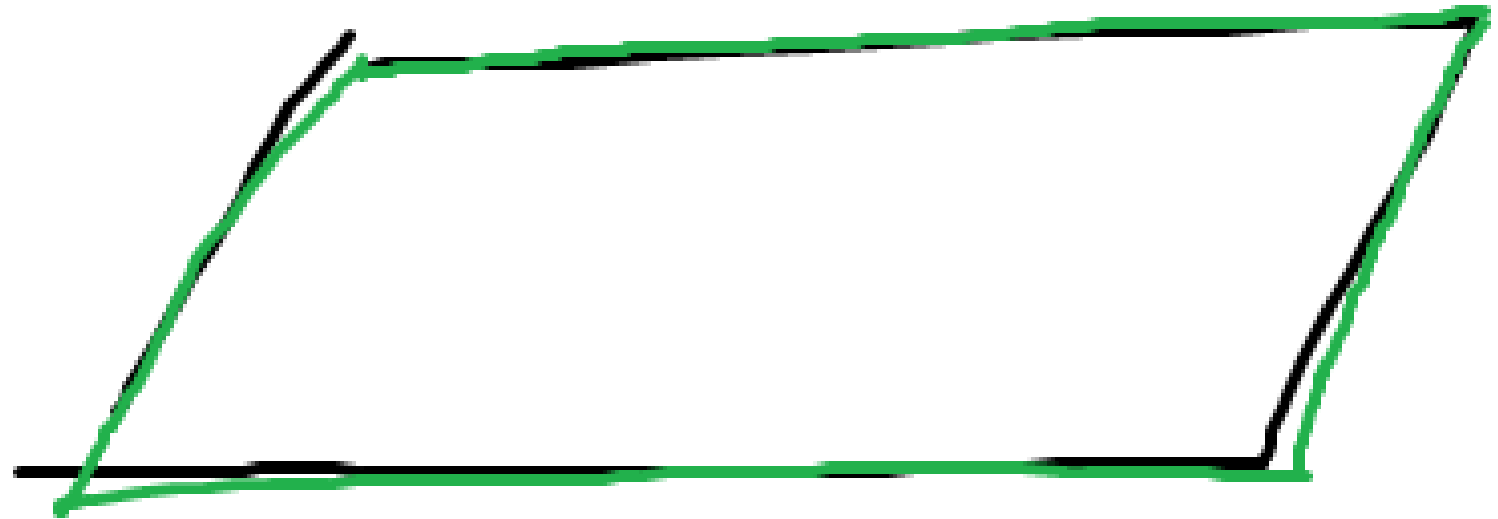
계속 매칭해보면서 최적의 값을 찾는 알고리즘



4. Ply 데이터 짚먹

Ply 데이터 핸들링 - Step 5 (RANSAC)

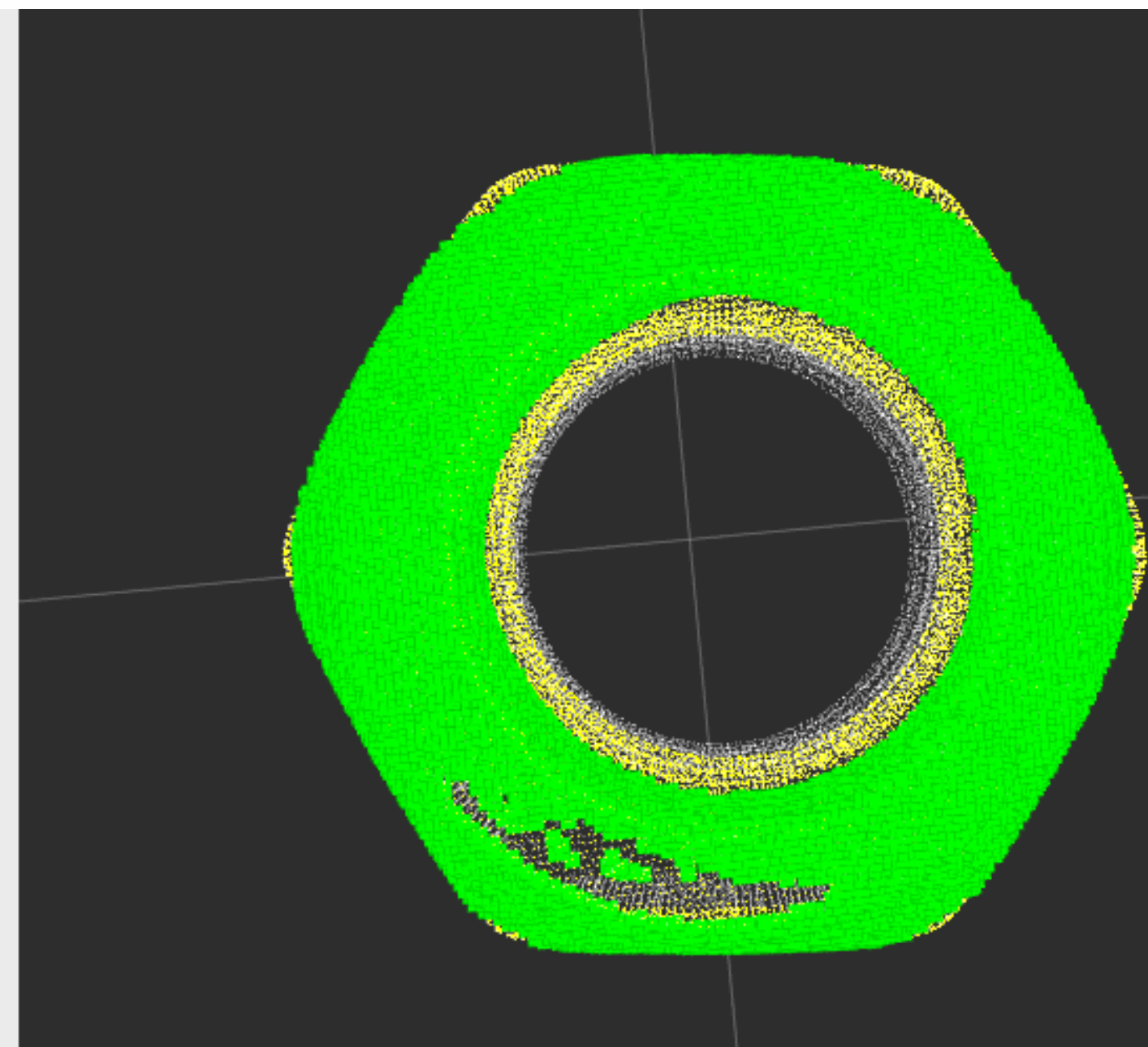
그럼 결국 최종 값은 다음과 같겠죠?



4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 5

Frame Rate	30
✓ Global Status: Ok	
✓ Fixed Frame	OK
Grid	☒
PointCloud2	☒
PointCloud2	☒
PointCloud2	☒
✓ Status: Ok	
Topic	/pointcloud_plane
Selectable	☒
Style	Squares
Size (m)	0.01
Size (Pixels)	10
Alpha	1
Decay Time	0
Position Transf...	XYZ
Color Transfor...	RGB8
Color	☐ 255; 255; 255
Axis	Z
Autocompute V...	☒
Use Fixed Frame	☒
Channel Name	intensity
Use rainbow	☒
Invert Rainbow	☐
Min Color	☒ 0; 0; 0
Max Color	☐ 255; 255; 255



4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 6

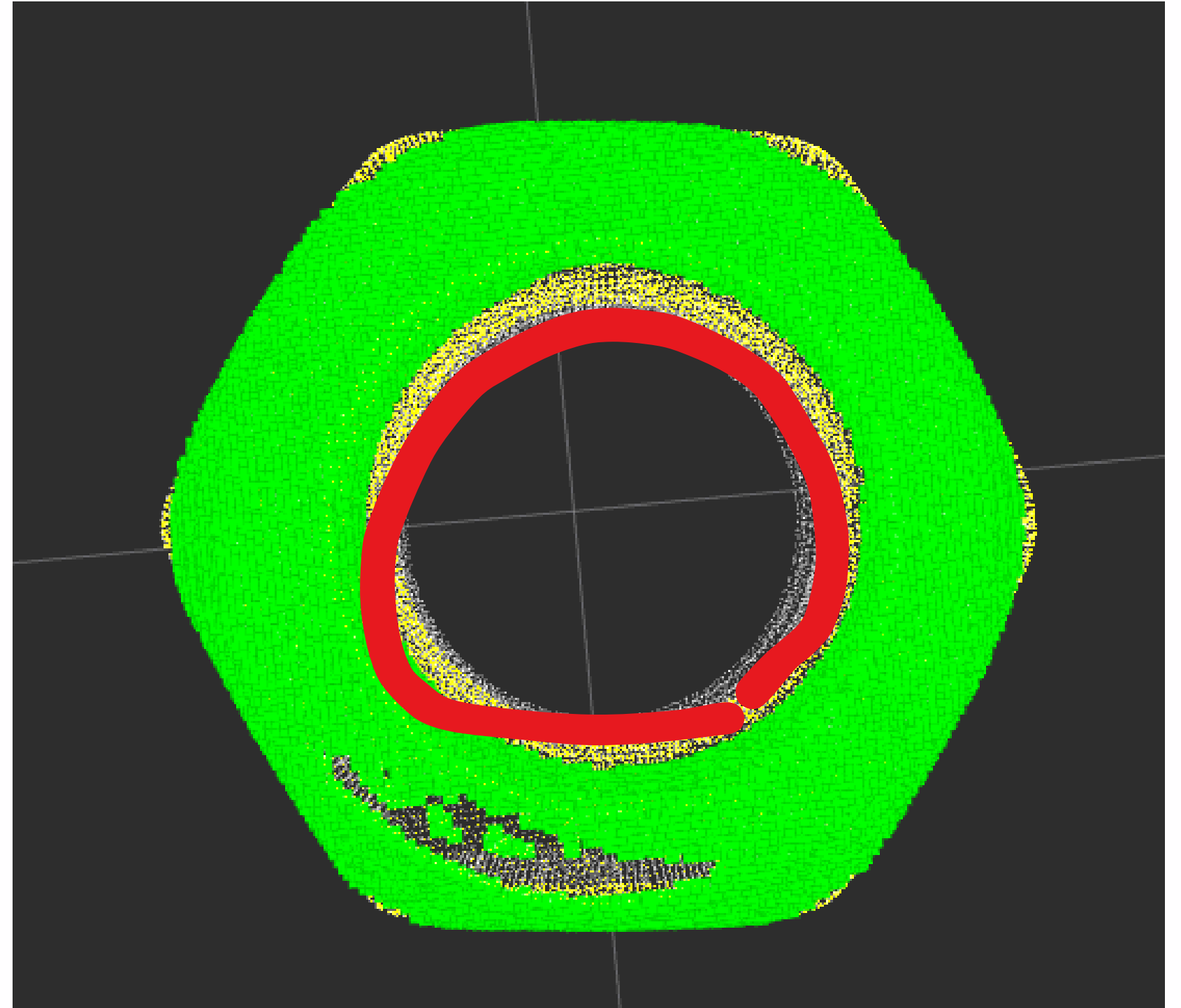
앞에서 성공적으로 찾았다면 외곽선 추출

```
// 6. 외곽선 추출 (평면 포인트에서 추출) ( 빨간색 )
CloudPtr boundary_cloud;
std::vector<pcl::Vertices> polygons;
extract_boundary_cloud(plane_cloud, boundary_cloud, polygons);
change_pointcloud_color(boundary_cloud, 255, 0, 0);
publishPointCloud(boundary_cloud, boundary_publisher_);
```

4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 6

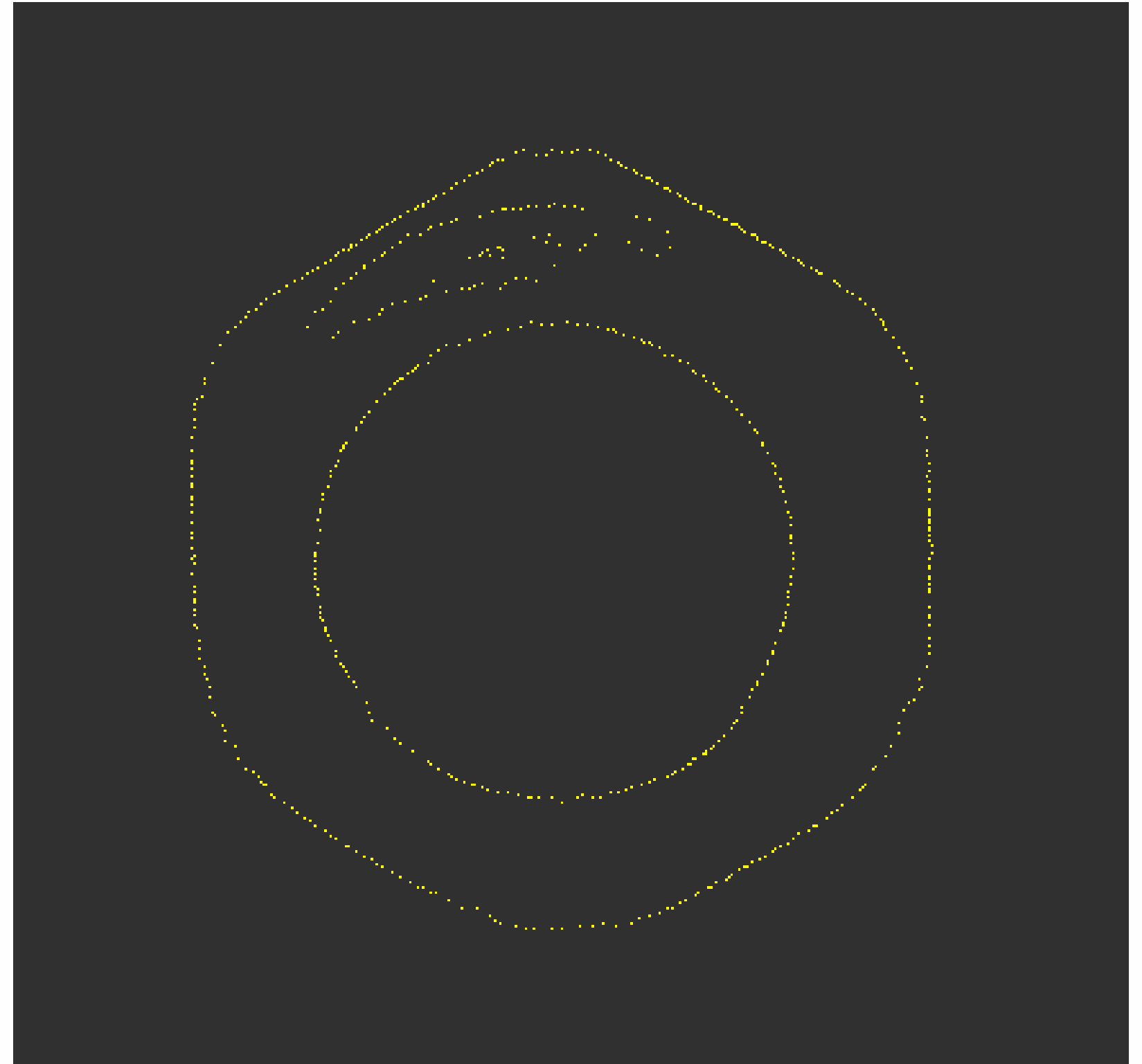
기대하는 값은 빨간색 라인!



4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 6

하지만 외곽선을 추출하기 때문에
다음과 같이 기대하는 값은
나오지 않아요



4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 7

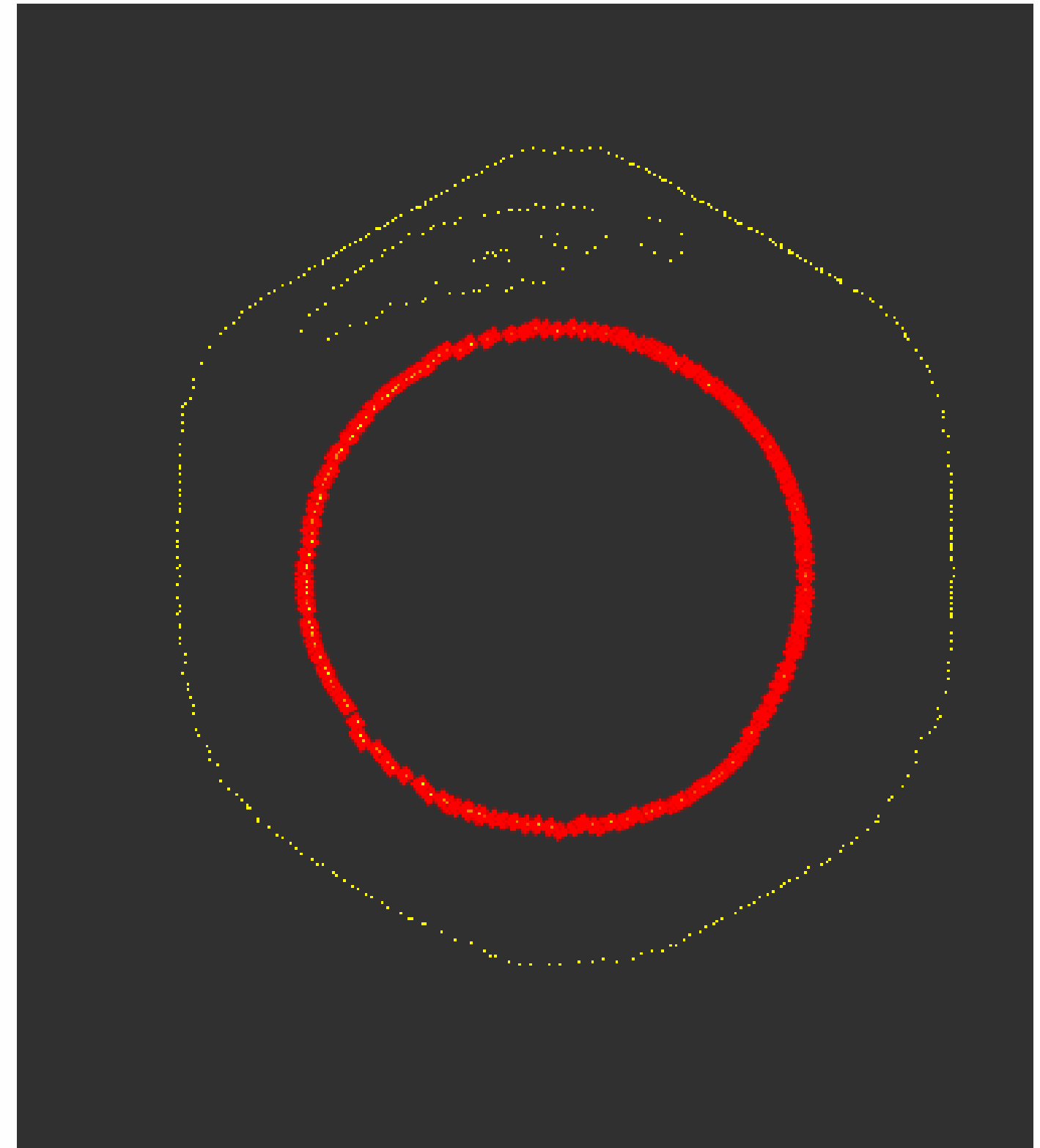
```
// 7. 찾은 외곽 포인트들 중 가장 작은 포인트 클라우드를 중심 포인트 클라우드로 설정 ( 파란색 )  
CloudPtr center_hole_cloud;  
get_center_hole(boundary_cloud, polygons, center_hole_cloud);  
publishPointCloud(center_hole_cloud, center_hole_publisher_);
```

4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 7

따라서 2번째로 큰 Pcl 데이터만
가져와서 출력하도록

PointCloud2	☒
▶ ✓ Status: Ok	
▶ Topic	/pointcloud_center_hole
Selectable	☒
Style	Squares
Size (m)	0.01



4. Ply 데이터 찍먹

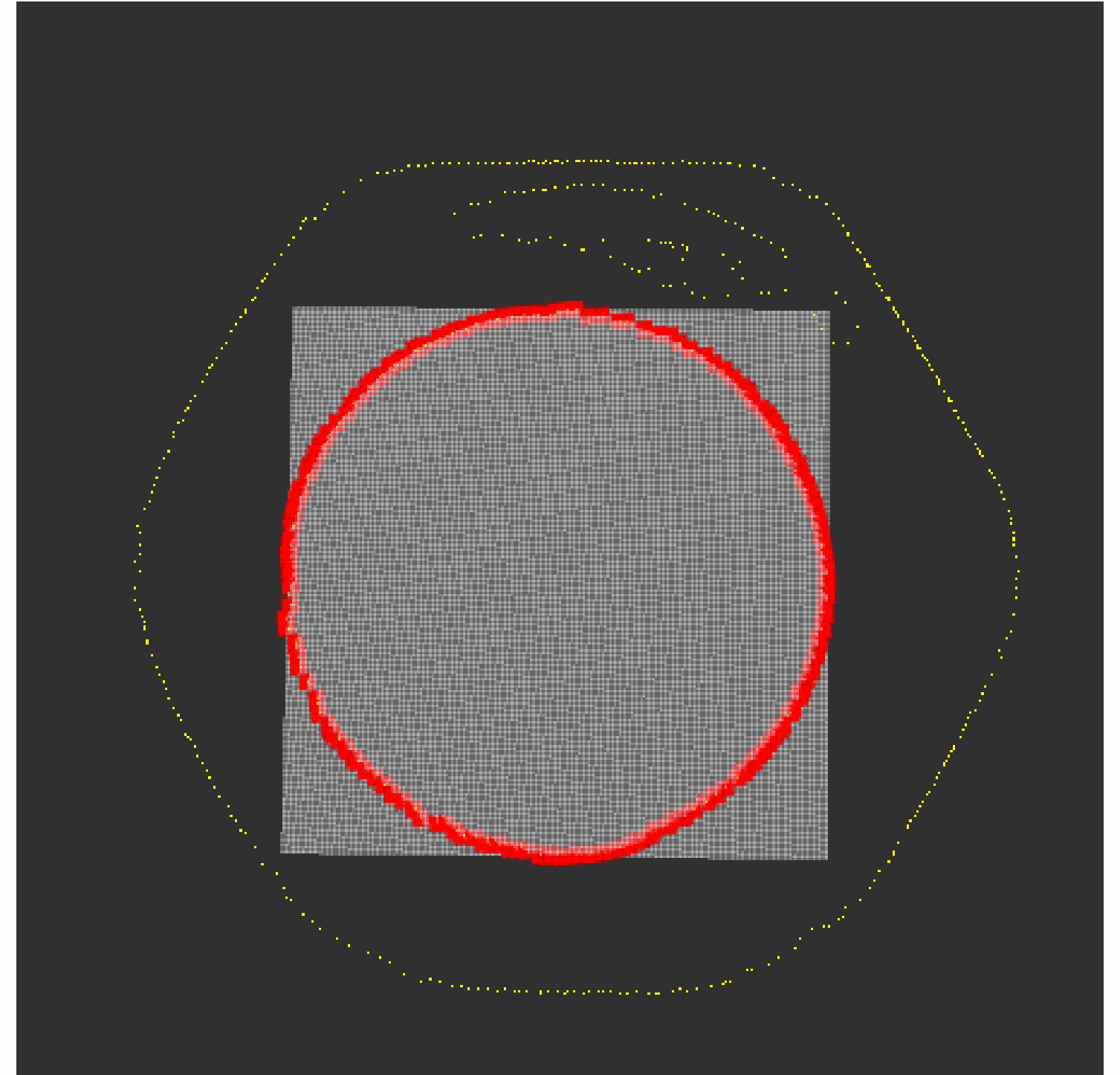
Ply 데이터 핸들링 - Step 8

```
// 8. 중앙 포인트 클라우드를 기준으로 사각형 생성 ( 흰색 )
pcl::PointCloud<pcl::PointXYZ>::Ptr square_cloud;
geometry_msgs::msg::TransformStamped tf_msg;
geometry_msgs::msg::TransformStamped up_tf_msg;
create_center_square_box(center_hole_cloud, "hole_center", square_cloud, tf_msg, up_tf_msg);
```

4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 8

 PointCloud2	☒
▶  Status: Ok	
▶ Topic	/pointcloud_center_square_box
Selectable	☒
Style	Points
Size (m)	0.01
Size (Pixels)	1
Alpha	0.1



4. Ply 데이터 찍먹

Ply 데이터 핸들링 - Step 8

TF	☒
Status: Ok	
Show Names	☐
Show Axes	☒
Show Arrows	☒
Marker Scale	1
Update Interval	0
Frame Timeout	9999
Frames	
All Enabled	☐
hole_center	☒
map	☐
up_hole_center	☐
Tree	
map	

